
Desarrollo de una herramienta en Python para el postprocesado y comparación de múltiples casos transitorios pertenecientes a un modelo térmico diseñado en **ESATAN-TMS**

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Contents

1. Abstract	1
2. Introduction	3
2.1. Objectives	3
3. State of the Art	5
3.1. Post-Processing Capabilities in ESATAN-TMS	5
3.2. Steady-State Thermal Model: HELIOS	6
3.3. Python-Based Post-Processing of Steady-State Cases	7
3.4. Motivation for the Present Work	8
4. Theoretical Background	9
4.1. Spacecraft Thermal Control	9
4.2. Space Thermal Environment	10
4.3. Heat Transfer and Thermal Energy Storage	11
4.3.1. Conduction Heat Transfer	11
4.3.2. Radiation Heat Transfer	13
4.3.3. Thermal Capacitance and Transient Response	16
4.4. Mathematical Modelling of Spacecraft Thermal Systems	17
4.4.1. Spatial Discretisation Approaches	17
4.4.2. Lumped Parameter Method (LPM)	17
4.4.3. Steady-State and Transient Formulations	19
4.4.4. Periodic Thermal Response and Cyclic Convergence	19
4.5. Thermal Modelling and Solution Framework in ESATAN-TMS	20
4.5.1. Geometrical and Radiative Models	20
4.5.2. Thermal Network Representation	20
4.5.3. Implementation of Steady-State, Transient and Cyclic Solutions	21

5. Thermal Model Development and Analysis Cases	23
5.1. Model Development and Geometrical Configuration	23
5.2. Thermal Model Hierarchy	25
5.3. Material and Surface Properties	26
5.3.1. Bulk Properties	27
5.3.2. Thermo-optical Properties	28
5.4. Nodal Discretisation and Thermal Network Definition	33
5.5. Orbital Environment and Radiative Cases	33
5.6. Transient Analysis Cases	35
5.6.1. Definition of Analysis Cases	35
5.6.2. Solver Configuration and Temporal Sampling	35
5.7. Thermal Model Verification	36
5.8. Simulation Results Export in TMD Format	36
6. Multi-Case Post-Processing Tool	39
6.1. Evolution of the Original Post-Processing Application	39
6.2. Application Architecture and Data Flow	41
6.3. Multi-Case Loading and Case Organisation	43
6.4. Functional Adaptation of the Application Interface	44
6.4.1. Time Selection	45
6.4.2. Multi-Case Representation Options	46
6.5. Saving, Restoring and Exporting Analyses	47
7. Transient Thermal Analysis	51
7.1. Temperature and Thermal-Load Evolution	51
7.2. Interface Temperature-Difference Analysis	54
7.3. Temperature Variation over Selected Time Windows	56
7.4. Transient Thermal Summary Table	59
7.5. Conductive and Radiative Heat-Flow Evolution	62
7.6. Absorbed and Released Thermal Energy	64
8. Electrical Power Balance and Battery Model	67

8.1. Solar-to-Electrical Power Conversion	67
8.1.1. Conversion Method	67
8.1.2. Input Selection and Electrical-Balance Display	69
8.2. Battery Model	70
8.2.1. Energy-Discharge-Level Basis	70
8.2.2. Thermal Derating and Battery-Pack Configuration	72
8.2.3. Electrical Parameters and CC–CV Charge Control	73
8.2.4. Battery Output Curves and Summary Results	75
9. Multi-Case Results and Discussion	79
9.1. Comparison Basis and Verification	79
9.2. Transient Thermal Comparison	79
9.2.1. Temperature and Load Evolution	79
9.2.2. Temperature Changes and Interface Differences	79
9.2.3. Heat Flows and Integrated Energy	79
9.3. Heater Influence in the Cold Case	79
9.4. Electrical and Battery Comparison	79
9.4.1. Solar Generation and Load Balance	79
9.4.2. Battery State and Operating Limits	79
9.5. Main Findings	79
10. Conclusions and Future Work	81
10.1. Conclusions	81
10.2. Future Work	81
References	83
A. Application Launch and Interface Overview	85
A.1. Inspection and Conductor Modules	85

List of Figures

3.1. External and internal views of the HELIOS thermal model. Taken from [1].	6
3.2. Post-processing architecture used in the previous project. Taken from [1].	7
4.1. Main radiative exchanges between an orbiting spacecraft and the external environment. Taken from [2].	10
4.2. One-dimensional heat conduction through a plane wall. Taken from [3].	12
4.3. Blackbody spectral emissive power at different temperatures and displacement of the emission maximum. Taken from [4].	14
4.4. Wavelength dependence and geometrical definition of thermal radiation.	15
4.5. Temperature approximation and conductive coupling in a lumped-parameter model. Taken from [2].	18
5.1. Internal configuration of the spacecraft thermal model viewed from opposite sides. The colours correspond to the geometry layer and do not represent temperatures. . .	24
5.2. Final external configuration of the spacecraft thermal model in ESATAN-TMS. The colours correspond to the geometry layer and do not represent temperatures. . . .	24
5.3. Example of nodal numbering for the OBC and adjacent radiator surfaces, shown from opposite sides. The figure illustrates the node ranges listed in Table 5.1. . . .	26
5.4. Thermal conductivity distribution over the final spacecraft model, viewed from opposite sides.	28
5.5. Solar absorptivity distribution over the final spacecraft model, viewed from opposite sides.	29
5.6. Solar reflectivity distribution over the final spacecraft model, viewed from opposite sides.	30
5.7. Infrared emissivity distribution over the final spacecraft model, viewed from opposite sides.	31
5.8. Infrared reflectivity distribution over the final spacecraft model, viewed from opposite sides.	32
5.9. Nominal orbital configuration and the 24 positions used for the radiative calculation. The green segment represents the portion outside eclipse, while the red segment indicates the eclipse interval.	34

5.10. <i>Solution Control</i> execution block used to obtain cyclic convergence and calculate the final transient orbital period in ESATAN-TMS.	35
6.1. Representative heat-flow visualisations generated for the HELIOS steady-state case with the original application. Extracted from [1].	40
6.2. Main data flow and software architecture of the current post-processing application.	42
6.3. Multi-case loading interface with one shared hierarchy and multiple .TMD result files.	43
6.4. Final post-processing interface after loading three transient cases. The sidebar groups the shared inputs, while the main area provides access to the model-inspection, transient and electrical-analysis modules.	45
6.5. Output-time selector used by the instantaneous inspection modules when transient data are available.	46
6.6. Configuration of the case-difference representation.	46
6.7. Sequence used to save, restore and manage a generated output.	48
7.1. Configuration area for selecting cases, model elements, represented parameters, display units and graphical settings.	52
7.2. Minimum, mean and maximum temperatures of SOLAR_PANEL_1 for the nominal transient case.	53
7.3. Battery temperatures and eclipse-dependent internal load Q_I for the nominal transient case.	53
7.4. Configuration area for selecting cases, source and target groups, the interface calculation basis and the optional temperature-difference limit.	54
7.5. Hot-node and cold-node temperatures for the interface between OBC_BOX and RADIATOR in the nominal transient case. The vertical marker identifies the maximum temperature difference.	55
7.6. Configuration area for selecting the cases, model elements, time windows, representation method and optional temperature-variation limits.	57
7.7. Maximum and minimum temperature variations of BODY_PANEL over 10- and 30-minute windows in the nominal transient case.	58
7.8. Configuration area for selecting the cases and model elements included in the transient thermal summary.	59
7.9. Configuration area for selecting the cases, source and target elements, heat-flow parameters and display units.	62

7.10. Conductive, radiative and total heat flows between BATTERY and BODY_PLATE_INF in the nominal transient case. Negative values indicate heat leaving the battery. . . .	63
7.11. Configuration area for selecting the cases, model element, thermal-power parameters and units used in the energy calculation.	65
7.12. Albedo and direct-solar thermal powers and their cumulative energies for SOLAR_PANELS in the nominal transient case.	66
8.1. Configuration of the solar-to-electrical conversion and the electrical-load selection. .	69
8.2. Illustrative solar-panel electrical balance containing $Q_{S,elec}$ and Q_{load}	70
8.3. Battery-temperature reference, charge derating and cell-pack configuration.	73
8.4. Battery efficiencies, power limits, internal resistances and simplified CC–CV control parameters.	75
8.5. Illustrative battery state-of-charge history.	76
8.6. Illustrative terminal-voltage and charge/discharge-current histories.	76
8.7. Illustrative battery temperature, charge-derating and efficiency histories.	77
A.1. Flow Explorer interface for hierarchical heat-flow inspection and multi-case representation.	86
A.2. Case-information interface showing the datasets, groups and couplings available in the loaded simulation.	87
A.3. Group-attribute interface for inspecting the selected model elements at a specified output instant.	88
A.4. Linear-conductor interface with case, group and output-time controls.	89
A.5. Radiative-conductor interface with case, group and output-time controls.	90
A.6. Nodal-information interface for the selected hierarchy elements.	91
A.7. Group-study interface for the inspection of selected components and their thermal connections.	92

List of Tables

5.1. Thermal-model hierarchy and node identifiers.	25
5.2. Material properties used in the thermal model.	27
5.3. Thermo-optical coating properties used in the thermal model.	28
5.4. Internal heat loads applied in the selected transient cases.	33
5.5. Transient cases selected from the common thermal model.	35
5.6. Structural and temporal characteristics of the exported .TMD files.	37
6.1. Main software groups and their functional responsibilities.	42
7.1. Maximum interface temperature differences detected from OBC_BOX in the nominal transient case.	55
7.2. Temperature variations of BODY_PANEL over the selected time windows in the nominal transient case.	59
7.3. Global temperature extrema of the selected elements in the nominal transient case. .	61
7.4. Maximum and minimum spatial temperature ranges of the selected elements in the nominal transient case.	61
7.5. Extreme conductive, radiative and total heat-flow values for the BATTERY-BODY_PLATE_INF selection in the nominal transient case.	64
8.1. Battery thermal and pack-configuration inputs.	73
8.2. Battery electrical and CC-CV control inputs.	75

Acronyms

Attitude Determination and Control System (ADCS)	23, 33
European Space Agency Thermal Analysis Tool (ESATAN-TMS) I, II, , 1, 3, 5–7, 20, 23, 24, 34–37, 39, 51, 66, 67	
Finite-Element Method (FEM)	17
Indium Tin Oxide (ITO)	28
Lumped Parameter Method (LPM)	, 17
Multi-Layer Insulation (MLI)	9, 23, 25, 27, 28, 31
On-Board Computer (OBC)	23, 33
Optical Solar Reflector (OSR)	28, 29
Systems Improved Numerical Differencing Analyzer/Fluid Integrator (SINDA/FLUINT) . 3, 5	
Simcenter 3D Space Systems Thermal (SST)	3, 5
Thermal Control System (TCS)	9
Thermal Desktop (TD)	3, 5
THERMICA Suite (THERMICA)	3, 5

1. Abstract

This Bachelor's Thesis focuses on the development of a Python-based post-processing tool for the thermal analysis of space systems using results generated with ESATAN-TMS.

Building upon a previous study limited to the post-processing of steady-state radiative cases, this work extends the analysis to fully transient thermal simulations. In addition, the tool has been designed to handle and compare multiple simulation cases associated with the same thermal model, allowing a more complete assessment of the system response under different operating conditions and orbital scenarios.

Since ESATAN-TMS presents limitations regarding integrated data visualization and case-to-case comparison, this project implements custom post-processing routines in Python, using libraries such as NumPy, Pandas, and Matplotlib. These routines automate the extraction, organization, and visualization of nodal results, including temperature evolution curves, transient heat flows, and comparative plots between different cases.

The resulting methodology improves the interpretation of complex transient thermal results and provides a more flexible framework for evaluating the thermal performance of a system throughout realistic orbital thermal cycles. By enabling the simultaneous analysis of several cases belonging to the same model, the developed tool supports a more efficient and systematic validation of thermal behaviour in space applications.

2. Introduction

Thermal management is an essential part of spacecraft design, as the temperature of each component must remain within allowable limits throughout the mission.

Since both the space environment and the operating conditions of the spacecraft evolve across different mission phases, steady-state cases cannot fully represent the real thermal behaviour. Transient simulations are therefore needed to examine the time-dependent behaviour of temperatures and heat flows.

The design of a spacecraft thermal control system relies on numerical analysis tools to predict system behaviour under different mission and operating conditions. These tools are used to represent the spacecraft through a thermal model and to calculate variables such as nodal temperatures and heat flows. Some of the best-known tools include THERMICA, Thermal Desktop together with SINDA/FLUINT, Simcenter 3D Space Systems Thermal (SST) and ESATAN-TMS, which provides a specialised environment for the development and solution of spacecraft thermal models.

The amount of information generated by these simulations can become considerable, particularly when transient analyses or multiple cases of the same model are considered. In such situations, the results must be organised and represented in a way that allows their evolution to be examined and the effects of the differences between cases to be identified. Dedicated post-processing tools can therefore support the interpretation of the simulation results.

This Bachelor's Thesis builds on an existing Python-based post-processing tool initially intended for steady-state analysis. The work combines the adaptation of the tool to transient and multi-case analysis with the development of a representative spacecraft thermal model in ESATAN-TMS. Selected thermal results are also used to perform a simplified electrical power analysis and provide the inputs for a separate battery model. The specific objectives of the project are presented in the following section.

2.1. Objectives

The main objective of the project is to enable the existing application to process, analyse, and compare several transient cases associated with the same ESATAN-TMS thermal model within a common environment.

Reaching this objective will require the coordinated development of the thermal model and the post-processing tool. The ESATAN-TMS model will provide a consistent set of transient results, while the application will be adapted to manage and compare them.

The thermal results will also be used to estimate the electrical power generated by the solar panels

and to calculate the electrical power balance. These calculations will then provide the inputs for a separate simplified battery model.

Accordingly, the specific objectives of the project will be:

- Develop a representative spacecraft thermal model and define transient cases under different environmental and operating conditions.
- Adapt the existing application to load and manage several transient result files while retaining a common model hierarchy.
- Implement methods for representing and comparing temperatures, thermal loads, heat flows and exchanged thermal energy.
- Generate graphical outputs and summary tables, and allow the corresponding analyses and configurations to be saved, restored and exported.
- Estimate solar-panel electrical generation, calculate the electrical power balance and use the resulting quantities as inputs to a separate simplified battery model.
- Apply the complete analysis process to the selected transient cases and compare the thermal responses obtained under different environmental and operating conditions.

3. State of the Art

Spacecraft thermal analyses are commonly performed using specialised software tools, including THERMICA, Thermal Desktop (TD) coupled with SINDA/FLUINT, SST and ESATAN-TMS. [5–7]

The result files generated by ESATAN-TMS contain a considerable amount of data that must be organised, examined, and interpreted to evaluate the thermal behaviour of the model. Two thermal regimes are considered in the analysis. A steady-state analysis provides a time-independent thermal solution for a predefined set of boundary conditions, whereas a transient analysis describes the evolution of the calculated results throughout the simulated period [8, 9].

To facilitate the interpretation of these results, an interactive Python-based post-processing tool was developed as part of an earlier project [1]. The underlying Python code extracts and organises the results generated by ESATAN-TMS and presents them through numerical and graphical representations. The original implementation was tested primarily using steady-state results from HELIOS, a thermal model developed in ESATAN-TMS, so the original version focuses on the analysis of a single steady-state case. The present work extends the tool to process transient results and compare several cases associated with the same thermal model.

3.1. Post-Processing Capabilities in ESATAN-TMS

ESATAN-TMS supports the numerical solution of spacecraft thermal models under both steady-state and transient conditions. For each analysis case, the resulting dataset contains nodal temperatures, applied thermal loads, and conductive and radiative heat flows. The software provides numerical and graphical tools for examining these results [9, 10].

The structure of the exported information depends on the selected outputs and the type of analysis. Transient results associate each variable with successive output times, and heat-flow data identify the nodes or surfaces involved in each exchange. In every case, numerical identifiers must remain linked to the physical elements they represent [9, 10].

These datasets provide the information required to evaluate the thermal behaviour of the model. When transient analyses or several simulation cases are considered, however, direct inspection can become less practical. Relevant results must therefore be selected, organised, and compared consistently.

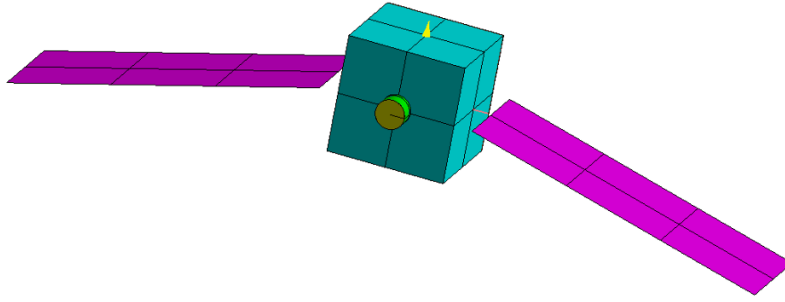
The post-processing functions available in ESATAN-TMS may not cover every requirement of a particular analysis. The Python implementation complements these functions by reorganising the exported thermal data according to an engineering hierarchy and providing additional methods for

selecting, grouping, and representing the results [9].

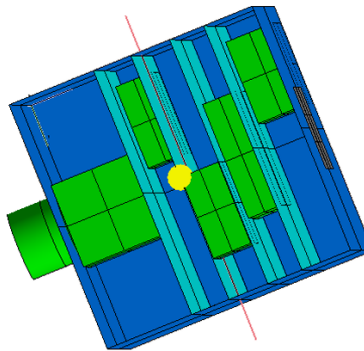
3.2. Steady-State Thermal Model: HELIOS

HELIOS is a thermal model developed in ESATAN-TMS with a single steady-state analysis case. The purpose of this model is to support the development and testing of the original Python-based tool. As a first approximation of a spacecraft thermal system, it provides sufficient detail to generate a consistent set of thermal results. The model contains the spacecraft structure, solar panels, batteries, radiating surfaces, and multi-layer insulation. The different components are connected through the conductive and radiative couplings that define the thermal network of the model [1].

The complete spacecraft configuration and a view of the internal components are shown in Figure 3.1.



(a) External view of the satellite model.



(b) Internal view of the satellite model.

Figure 3.1: External and internal views of the HELIOS thermal model. Taken from [1].

The steady-state solution of HELIOS contains the thermal data that are extracted and organised

by the original Python code before being examined through the graphical interface. In contrast, the present work uses a separately developed spacecraft thermal model with several cyclic transient analysis cases.

3.3. Python-Based Post-Processing of Steady-State Cases

The original Python-based implementation uses the steady-state results generated for the HELIOS model in ESATAN-TMS [1]. In this type of analysis, a single temperature is calculated for each node under the selected boundary conditions, while the corresponding thermal loads and heat flows remain independent of time.

The imported results are linked to the hierarchy defined for the thermal model, so that each node can be located within the corresponding component and subsystem. Instead of working directly with the complete list of node identifiers contained in the ESATAN-TMS files, the user can examine the results through the spacecraft structure. This relationship between the thermal data and the physical model helps identify the origin of each value and interpret it in an engineering context.

The main modules of the original codebase and their connections are shown in Figure 3.2.

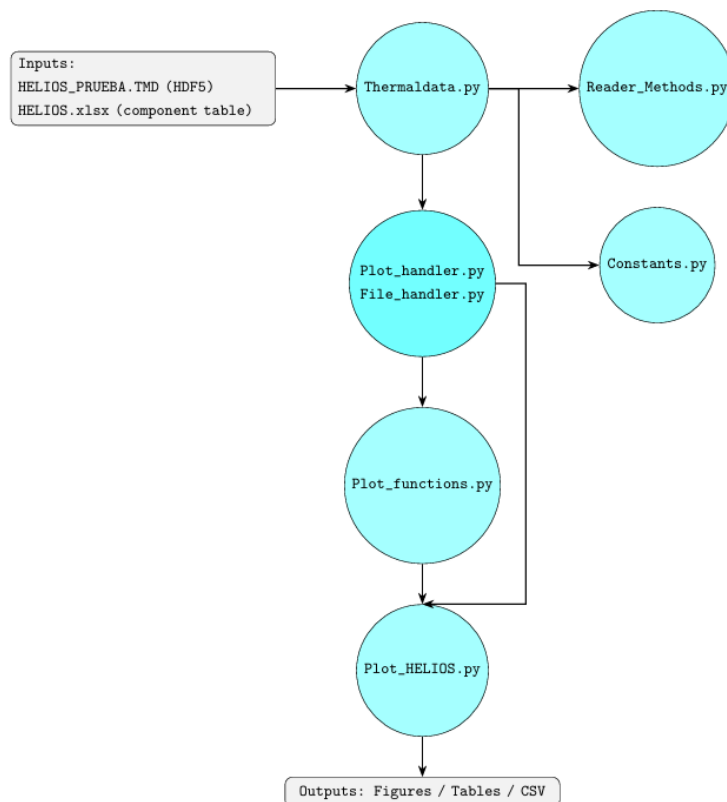


Figure 3.2: Post-processing architecture used in the previous project. Taken from [1].

The architecture separates result reading, data organisation and representation. `Thermaldata.py` imports the HELIOS result file and the external component table, while the supporting modules handle data selection, file management and plotting. The interface is provided by `Plot_HELIOS.py`, from which figures, tables and data files can be generated [1].

This version is suitable for examining the thermal results of an individual steady-state case. However, it does not provide complete post-processing of transient results or allow several simulation cases to be loaded and compared simultaneously. Extending the same structure to time-dependent results and multiple cases requires changes to both data management and representation.

3.4. Motivation for the Present Work

Transient simulations associate each calculated value with a specific simulation time. The post-processing tools must preserve this correspondence when results are selected, represented, and compared, particularly when several components or interfaces are considered together.

During an orbital cycle, transitions between solar illumination and eclipse, changes in internal dissipation, and the operation of active thermal-control elements produce variations in the thermal response of the spacecraft. These variations must therefore be examined throughout the orbital cycle to determine when they occur and which elements are most affected.

Comparing simulation cases introduces an additional requirement. Spacecraft thermal analyses may include several cases representing different environmental and operating conditions. Processing them separately requires equivalent elements to be selected again for every case and prevents their direct comparison within the same representation. Managing cases associated with the same thermal model together allows equivalent selections to be applied and the differences between them to be calculated consistently.

For these reasons, the existing implementation is extended to process transient results and manage several simulation cases associated with the same thermal model. This allows equivalent components and interfaces to be compared throughout the simulated period and the intervals in which the differences between cases are most significant to be identified.

4. Theoretical Background

Spacecraft temperatures are determined by the interaction between internal heat dissipation, external radiative loads and heat transfer through the structure. These contributions change throughout the mission, leading to different temperature conditions in each component. Their analysis requires an understanding of the space environment, the main heat-transfer mechanisms and the mathematical methods used to represent the physical system [2, 8, 11].

4.1. Spacecraft Thermal Control

Thermal control is required in many engineering systems because temperature affects equipment performance, material behaviour and reliability. Heat generated within a system and energy exchanged with the external environment must be managed so that the components remain within acceptable temperature limits [3].

In spacecraft, the Thermal Control System (TCS) maintains onboard components within their specified temperature limits throughout the mission, despite changes in environmental and operating conditions. Excessive temperatures, large thermal gradients and repeated thermal cycles can shorten component life, reduce measurement accuracy or prevent equipment from functioning correctly. The design must also satisfy the restrictions imposed by the available mass, volume and electrical power [2, 11, 12].

Thermal requirements depend on the function and operating state of each component. They include allowable temperature ranges, limits on thermal gradients and requirements for temperature stability. For example, electrical and electronic equipment, including batteries and onboard computers, operate within specified temperature ranges. Optical instruments such as telescopes often require tighter control of temperature gradients to limit dimensional changes and maintain alignment [11, 12].

The temperature of each component depends on the balance between internally dissipated power and the energy exchanged with the external environment. This balance is influenced by the amount of energy absorbed by the external surfaces, the heat transferred through the internal structure and the energy emitted towards space. Since these contributions change during the mission, the required control must remain effective under different operating and environmental conditions [2, 11].

Different thermal-control measures are combined to meet these requirements. Surface coatings modify solar absorption and infrared emission, radiators provide suitable surfaces for heat rejection, and Multi-Layer Insulation (MLI) limits unwanted radiative exchange. Conductive links distribute heat between components and structural elements, while heaters provide additional energy when

passive measures are insufficient under cold conditions [2, 11, 13].

The selection of these measures depends on the permitted temperature ranges, expected heat loads and mission phases. Thermal control is therefore not limited to maintaining a single uniform temperature. It must also control gradients, reduce excessive variations and meet the individual requirements of each component [11, 12].

4.2. Space Thermal Environment

The space environment is one of the main factors affecting heat exchange through the external surfaces of a spacecraft. In orbit, energy is exchanged with space mainly by radiation because external convective transport is absent. Conduction remains important within the structure and between connected components, but it does not provide a direct mechanism for rejecting heat to space [2, 11].

For a spacecraft orbiting the Earth, the main environmental heat inputs are direct solar radiation, solar radiation reflected by the Earth, known as albedo, and infrared radiation emitted by the planet. At the same time, the external surfaces emit thermal radiation towards deep space. These exchanges are represented in Figure 4.1 [2, 11, 13].

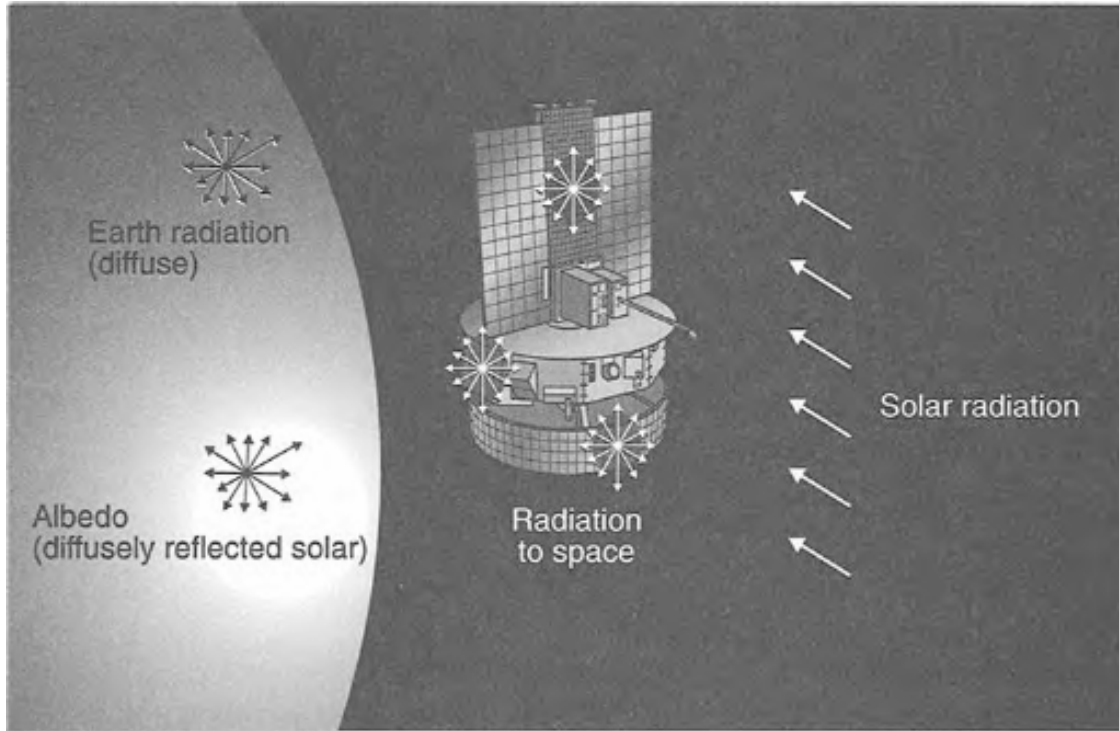


Figure 4.1: Main radiative exchanges between an orbiting spacecraft and the external environment. Taken from [2].

Direct solar radiation generally provides the largest environmental heat input for an Earth-orbiting spacecraft. The power absorbed by a flat surface depends on the solar irradiance, surface area, orientation and solar absorptivity. It can be expressed as [11]:

$$\dot{Q}_{\text{sun}} = \alpha_s G_s A \cos \theta \quad (4.1)$$

where α_s is the solar absorptivity, G_s is the solar irradiance, A is the surface area and θ is the angle between the surface normal and the solar rays. Since G_s is expressed in W m^{-2} , the resulting absorbed power is expressed in W [11].

Albedo is solar radiation reflected by the Earth's surface and atmosphere. The albedo's contribution depends on the illuminated portion of the planet visible from the spacecraft and on the fraction of reflected energy absorbed by each exposed surface. Planetary infrared radiation instead originates from the Earth's thermal emission and depends on the infrared properties of the exposed surfaces. In both cases, the incoming energy also depends on the portion of the planet visible from each surface [2, 11].

The contribution made by each source varies with the orbit, spacecraft attitude, exposed area and thermo-optical properties. During a total eclipse, the direct-solar term becomes zero, and rapid variations occur when the spacecraft enters or exits the eclipse region. Albedo follows the visible illuminated region of the Earth, whereas planetary infrared radiation remains present during eclipse [11, 13].

The periodic sequence of illuminated and eclipse phases exposes the spacecraft to alternating heating and cooling during each orbit. Component temperatures adjust over a finite interval because energy is stored and transferred through the structure. A time-dependent analysis is required to represent this delayed response to the changing environmental loads [2, 11].

4.3. Heat Transfer and Thermal Energy Storage

The temperature response of a spacecraft depends on the amount of heat transferred and stored over time. Heat transfer occurs by conduction, convection or radiation. Conduction and radiation are the main mechanisms in spacecraft thermal analysis, while thermal capacitance accounts for the energy stored by each component [3, 11].

4.3.1. Conduction Heat Transfer

Conduction transfers thermal energy through a material or between bodies in contact without bulk movement of matter. The conductive heat flow depends on the temperature gradient, the thermal conductivity of the material and the area normal to the direction of transfer. For one-dimensional

conduction, Fourier's law is written as [3,4]

$$\dot{Q}_x = -kA \frac{dT}{dx}, \quad (4.2)$$

where $\dot{Q}_x$ is the conductive heat flow, k is the thermal conductivity, A is the area normal to the heat flow and dT/dx is the temperature gradient. The negative sign indicates that heat flows in the direction of decreasing temperature.

Figure 4.2 shows the one-dimensional plane-wall configuration described by Fourier's law.

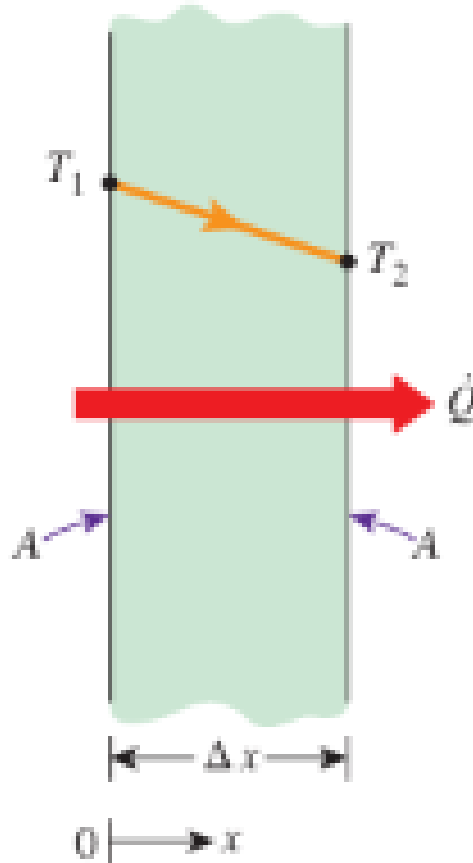


Figure 4.2: One-dimensional heat conduction through a plane wall. Taken from [3].

For a plane layer of thickness L , constant area and constant conductivity, Fourier's law can also be

expressed in terms of thermal resistance [3, 4]:

$$R_{\text{cond}} = \frac{L}{kA} \quad (4.3)$$

$$\dot{Q}_{12} = \frac{T_1 - T_2}{R_{\text{cond}}} \quad (4.4)$$

The thermal-resistance formulation allows conductive paths to be combined in series or parallel. At an interface between two components, the effective resistance also depends on surface roughness, contact pressure and any intermediate materials. These effects are represented by the contact resistance, while the linear thermal conductance is defined as the inverse of that resistance [2, 3]:

$$G_{12}^L = \frac{1}{R_{12}} \quad (4.5)$$

where G_{12}^L is the linear thermal conductance between components 1 and 2, and R_{12} is the corresponding resistance. This relation is particularly useful in nodal models because each conductive connection is represented by a conductance between two nodes. When temperature changes with position or time, Fourier's law is combined with conservation of energy to obtain the heat-diffusion equation [3, 4, 8].

4.3.2. Radiation Heat Transfer

Unlike conduction, thermal radiation transfers energy through electromagnetic waves, which propagate through vacuum. Radiative exchange takes place between spacecraft surfaces, while external surfaces also emit energy towards space. The power emitted by a real surface depends on the absolute temperature, area and emissivity, as expressed by the Stefan–Boltzmann law [3, 11]:

$$\dot{Q} = \varepsilon \sigma A T^4 \quad (4.6)$$

where ε is the surface emissivity, σ is the Stefan–Boltzmann constant, A is the emitting area and T is the absolute surface temperature. The value of the Stefan–Boltzmann constant is $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ [3].

For an external surface facing deep space without obstruction, the net radiative heat flow is expressed as [11]:

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A (T^4 - T_{\text{space}}^4) \quad (4.7)$$

where T_{space} is the effective temperature assigned to deep space. The term T_{space}^4 is generally small because the deep-space temperature is much lower than the spacecraft surface temperature [11].

The Stefan–Boltzmann law gives the total power emitted over all wavelengths. The spectral distribution, however, depends on temperature. As temperature increases, the emitted power rises and the maximum moves towards shorter wavelengths. This behaviour explains why radiation received

from the Sun is mainly treated as short-wave radiation, whereas the Earth and spacecraft surfaces emit primarily within the infrared range, as illustrated in Figure 4.3 [3, 11].

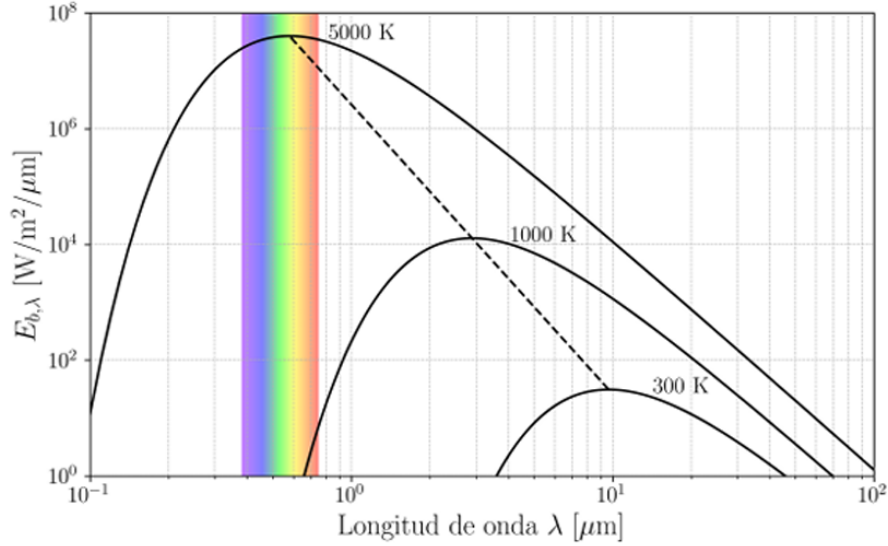


Figure 4.3: Blackbody spectral emissive power at different temperatures and displacement of the emission maximum. Taken from [4].

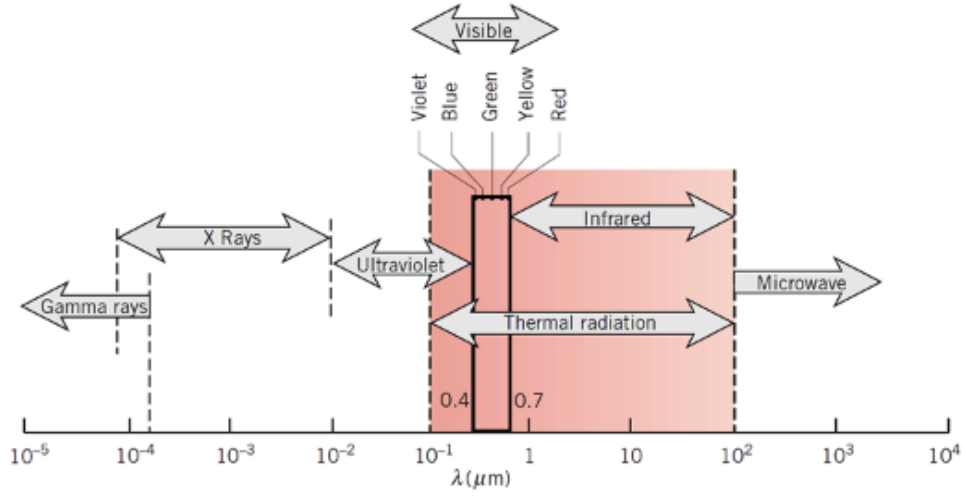
When radiation reaches a surface, part of it is absorbed, part is reflected and the remainder, if any, is transmitted. Conservation of radiative energy requires the corresponding fractions to satisfy [3]

$$\alpha + \rho + \tau = 1 \quad (4.8)$$

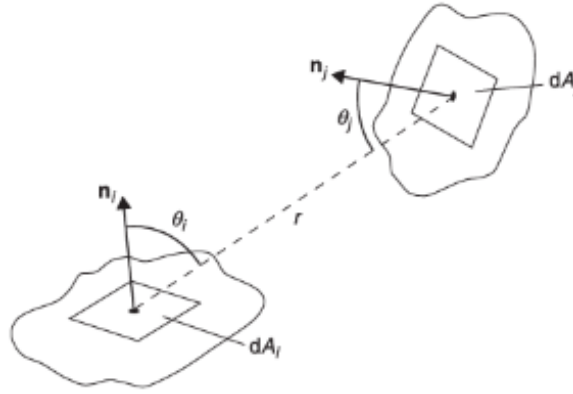
where α , ρ and τ are absorptivity, reflectivity and transmissivity, respectively. For an opaque surface, no radiation is transmitted, and Equation 4.8 reduces to $\alpha + \rho = 1$ [3, 11].

Since these properties depend on wavelength, the solar and infrared bands are treated separately in spacecraft thermal analysis. Solar absorptivity represents the fraction of short-wave solar radiation absorbed by a surface, while infrared emissivity governs the exchange of long-wave thermal radiation. The same surface can therefore have different properties in the two bands [11, 13, 14]. The wavelength range associated with thermal radiation is shown in Figure 4.4a.

The view factor F_{ij} represents the fraction of radiation leaving surface i that reaches surface j . The value of F_{ij} depends on the relative position, orientation and area of the two surfaces, as illustrated in Figure 4.4b [11].



(a) Wavelength range of thermal radiation. Taken from [14].



(b) Geometry used to define a view factor. Taken from [11].

Figure 4.4: Wavelength dependence and geometrical definition of thermal radiation.

View factors satisfy the reciprocity relation. When a set of N surfaces forms a closed enclosure, the factors from each surface also satisfy the following rule [3, 11]:

$$A_i F_{ij} = A_j F_{ji} \quad (4.9)$$

$$\sum_{j=1}^N F_{ij} = 1 \quad (4.10)$$

where N is the number of surfaces forming the enclosure.

Radiative exchange depends on the surface temperatures and thermo-optical properties, as well as on the areas and relative geometry of the surfaces. Analytical view factors are available for simple

arrangements, whereas spacecraft radiative models generally calculate them numerically [11, 13].

4.3.3. Thermal Capacitance and Transient Response

Thermal capacitance measures the energy required to change the temperature of a body. For a body with uniform temperature and constant properties, it is given by [2, 3]:

$$C = mc_p \quad (4.11)$$

$$\Delta E = C\Delta T \quad (4.12)$$

where C is the thermal capacitance, m is the mass, c_p is the specific heat capacity and ΔE represents the changes in the stored thermal energy of a body.

For a transient process, the energy balance is expressed as [11]:

$$C \frac{dT}{dt} = \dot{Q}_{\text{net}} \quad (4.13)$$

where $\dot{Q}_{\text{net}}$ is the net heat flow into the body. For the same net heat flow, a larger thermal capacitance results in a slower temperature change [3].

For a single body connected to a constant-temperature boundary through a linear thermal resistance R , the thermal time constant and the corresponding temperature response are expressed as [3]:

$$\tau_{\text{th}} = RC \quad (4.14)$$

$$T(t) = T_{\infty} + (T_0 - T_{\infty}) \exp\left(-\frac{t}{\tau_{\text{th}}}\right) \quad (4.15)$$

where T_0 is the initial temperature and T_{∞} is the equilibrium temperature approached as time increases. A larger time constant corresponds to a slower response [3].

During a transient process, heat flow changes with time, and the net energy transferred between two instants is obtained by integrating the heat flow over the corresponding interval [3]:

$$E_{12} = \int_{t_1}^{t_2} \dot{Q}(t) dt \quad (4.16)$$

The sign of E_{12} follows the sign convention adopted for $\dot{Q}$. In orbital analysis, this integral quantifies the energy absorbed or released during a selected part of the orbit. For example, energy is stored during an illuminated interval and released after eclipse entry. The corresponding temperature change is delayed by the thermal capacitance by the thermal capacitance and the conductive and radiative connections of the component with the rest of the system [2, 11].

4.4. Mathematical Modelling of Spacecraft Thermal Systems

Spacecraft thermal models combine the equations governing heat transfer and energy storage. The principle of conservation of energy is applied to calculate temperatures and heat flows under specified loads and boundary conditions. Analytical methods provide exact solutions when the geometry and boundary conditions are simple. More complex configurations, involving multiple materials, interfaces and time-dependent inputs, are formulated as a finite system of equations for numerical analysis [11, 13].

4.4.1. Spatial Discretisation Approaches

The numerical solution of heat-transfer equations requires a discrete representation of the geometry and temperature field. The geometry is divided into a finite number of regions, each associated with one or more temperature values. This process is known as spatial discretisation [11, 13].

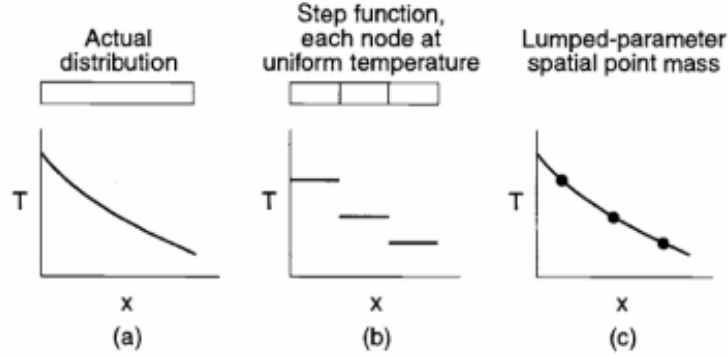
Several numerical methods are available to discretise the thermal problem. Finite-difference methods express spatial derivatives through temperature differences between adjacent grid points. The Finite-Element Method (FEM) divides the geometry into connected elements and describes the temperature variation within each element using interpolation functions. Finite-volume methods apply an energy balance to each control volume using the heat flows across the control-volume boundaries. The three formulations differ, but all of them reduce the thermal problem to a finite system of equations expressed in terms of discrete temperatures [2, 13].

At system level, spacecraft thermal models commonly use a network of lumped nodes. Each node represents a particular region of the model with a single temperature value. Regions where small temperature gradients are expected can be represented by one node, whereas regions with larger gradients or relevant heat-transfer paths are divided into several nodes [11, 13].

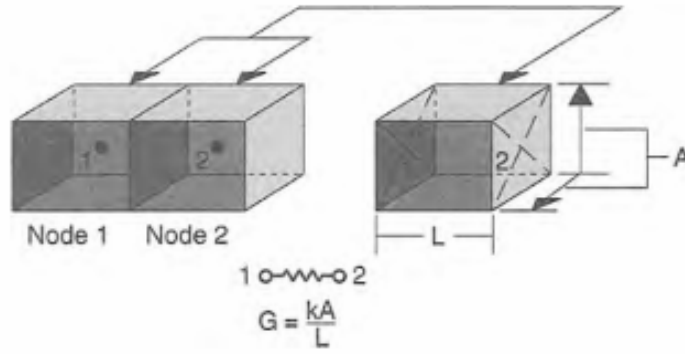
A finer division provides greater spatial resolution, but it also increases the number of conductive and radiative couplings, the calculation time and the volume of output data. The selected nodalisation must resolve the temperatures, gradients and heat paths relevant to the analysis without introducing unnecessary detail [2, 13].

4.4.2. Lumped Parameter Method (LPM)

In the LPM, a thermal system is represented as a network of discrete nodes linked by conductive and radiative couplings. Each node represents a region whose temperature is assumed to be spatially uniform, while the energy stored in that region is represented by a thermal capacitance. Figure 4.5 illustrates the approximation of a continuous temperature distribution by discrete nodal values and the conductive connection between two adjacent nodes [2, 8].



(a) Approximation of a continuous temperature distribution by discrete nodal values.



(b) Conductive connection between two adjacent nodes.

Figure 4.5: Temperature approximation and conductive coupling in a lumped-parameter model. Taken from [2].

The principle of conservation of energy is applied to each of the N capacitive nodes to establish the corresponding energy balance. This converts the continuous temperature field into a coupled system of N first-order ordinary differential equations, one for each nodal temperature. The corresponding balance for a general node i is written as [8, 11]:

$$C_i \frac{dT_i}{dt} = \dot{Q}_i(t) + \sum_{\substack{j=1 \\ j \neq i}}^N G_{ij}^L (T_j - T_i) + \sigma \sum_{\substack{j=1 \\ j \neq i}}^N G_{ij}^R (T_j^4 - T_i^4) \quad (4.17)$$

where C_i is the nodal thermal capacitance, $\dot{Q}_i(t)$ is the net heat load applied to the node, G_{ij}^L is the linear conductance, G_{ij}^R is the radiative exchange coefficient, σ is the Stefan–Boltzmann constant and N is the total number of nodes in the network.

Conductive exchange is proportional to the temperature difference $T_j - T_i$, whereas radiative exchange depends on $T_j^4 - T_i^4$. This T^4 dependence makes the resulting system non-linear [8, 11].

The nodal equations are completed by assigning the applied loads and boundary conditions. The loads represent internally dissipated power, external radiative heating or heat supplied by temperature-controlled devices. Boundary conditions specify temperatures, heat inputs or other interactions that are not calculated as unknown nodal temperatures [8, 10].

A transient calculation also requires an initial temperature for each capacitive node. This initial distribution affects the beginning of the solution and is obtained from an estimated condition, a previous transient interval or a steady-state calculation. The influence of this initial distribution must be considered when the required result is a periodic orbital response [8, 10].

4.4.3. Steady-State and Transient Formulations

In a steady-state solution, the nodal temperatures remain constant. As a result, dT_i/dt is zero, and the storage term in Equation (4.17) disappears. The resulting non-linear algebraic equations determine the steady-state temperatures for fixed loads and boundary conditions. However, they contain no information about the transition towards this equilibrium state [8, 11].

In a transient solution, the storage term is retained and an initial temperature is assigned to each capacitive node. The coupled differential equations are integrated over time, allowing the loads and boundary conditions to change during the selected interval. The resulting solution describes temperatures and heat flows as functions of time and includes the energy stored in the capacitive nodes [8, 11].

A transient calculation can therefore identify temporary maximum and minimum temperatures, the time at which they occur and the delay between a change in load and the corresponding temperature response. The selected formulation depends on whether only an equilibrium condition is required or whether the evolution of the system must also be represented [2, 13].

4.4.4. Periodic Thermal Response and Cyclic Convergence

When the orbital geometry and operating sequence repeat after a period P , the applied loads are also periodic. The thermal response is periodic when the nodal temperatures repeat after the same period. To approach this condition, the final temperature distribution from each calculated cycle is used to initialise the next one [10, 11, 15].

$$\dot{Q}_i(t + P) = \dot{Q}_i(t) \quad (4.18)$$

$$T_i(t + P) = T_i(t) \quad (4.19)$$

In numerical calculations, small differences generally remain between the temperatures calculated at the beginning and end of a complete period. Cyclic convergence is accepted when the maximum

of these differences is below a selected tolerance [8, 10].

$$\max |T_i(t_0 + P) - T_i(t_0)| \leq \varepsilon_{cyc} \quad (4.20)$$

where ε_{cyc} is the convergence criterion. Once this condition is satisfied, the calculated cycle represents the response to the periodic loads rather than the initial numerical adjustment of the model [8, 10, 15].

4.5. Thermal Modelling and Solution Framework in ESATAN-TMS

ESATAN-TMS applies a lumped-parameter nodal formulation through connected geometrical, radiative and thermal models. The physical description is converted into a network containing thermal capacitances, conductive and radiative couplings, boundary conditions and applied loads [8, 9, 16].

The modelling process links the physical configuration with the equations solved during the analysis. Geometrical and surface information is used in the radiative calculations, while material properties and physical connections determine thermal storage and conduction. These contributions are combined in the thermal network before the selected numerical solution is performed [8, 9].

4.5.1. Geometrical and Radiative Models

The geometrical model defines the dimensions, positions and orientations of the modelled elements, together with their materials and thermo-optical properties. The geometrical discretisation also establishes the regions that can later be associated with individual thermal nodes [9, 16].

The corresponding surfaces form the radiative model. View factors and radiative exchange factors between surfaces are calculated from their geometry and properties. During an orbital analysis, the model also determines the solar, albedo and planetary infrared loads absorbed at the selected orbital positions [9, 13, 16].

4.5.2. Thermal Network Representation

The thermal network consists of capacitive nodes linked by conductors. Each capacitive node represents a region with an associated temperature and heat-storage capacity. The nodal capacitance is calculated from the mass and specific heat assigned to the corresponding part of the model [8, 9].

Linear conductors represent solid conduction, contact conductance and other couplings proportional to temperature difference. Radiative conductors represent energy exchange between surfaces and are obtained from the radiative calculation. Environmental inputs, internal dissipation and boundary conditions complete the nodal energy balances [8–10].

Nodes and conductors are generated from the geometry or defined explicitly when a physical element or connection is not represented directly by the geometrical model. This allows equipment units, dedicated conductive links and user-defined thermal interactions to be included without adding unnecessary geometrical detail [8,9].

4.5.3. Implementation of Steady-State, Transient and Cyclic Solutions

The steady-state solver determines the nodal temperatures for fixed loads and boundary conditions. The transient solver retains the nodal capacitances and initial temperatures and integrates the energy-balance equations over time. Calculation and output times are selected according to the temporal resolution required for the analysis [8,10].

For periodic cases, the cyclic solver repeats the defined interval and uses the final temperature distribution of one cycle to initialise the next. The process continues until the difference between consecutive cycles satisfies the selected convergence criterion or the permitted number of cycles is reached [8,10,16].

The numerical solution provides the nodal temperatures and the contributions included in their energy balances. Depending on the selected outputs, these results may include applied loads and conductive and radiative heat flows, which can subsequently be examined as steady values or as functions of time [9,10].

5. Thermal Model Development in ESATAN-TMS and Transient Analysis Cases

The thermal model developed in ESATAN-TMS generates the results analysed throughout this work. It does not reproduce a particular flight design in full detail. Instead, it represents the main heat-transfer interactions with sufficient geometrical and physical detail to generate transient results and compare different cases. The same geometry, node hierarchy and thermal network are retained in every simulation, while the environmental and operating conditions are modified between cases.

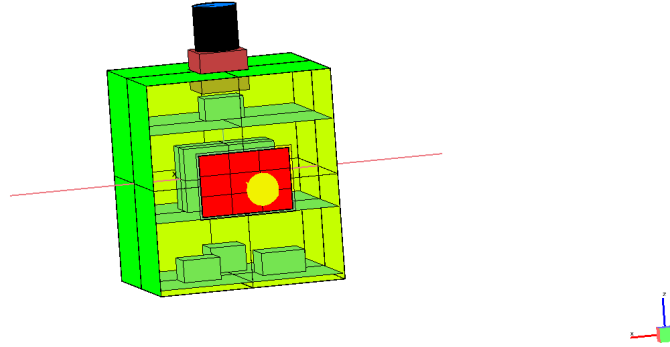
The model contains the main body, two solar-panel wings, a camera, several internal electronic units, a battery, a radiator and Multi-Layer Insulation (MLI). Conductive and radiative couplings connect these elements, allowing temperatures and heat flows to be calculated throughout each simulation. The model follows the lumped-parameter formulation presented in Chapter 4 and implemented in ESATAN-TMS [8,9].

5.1. Model Development and Geometrical Configuration

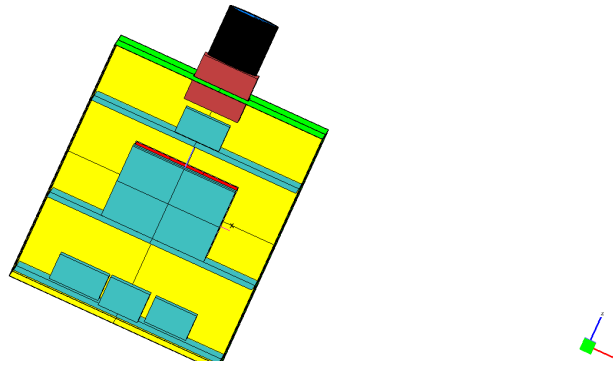
The geometrical model is refined to represent the arrangement of the internal plates and equipment and the main conductive connections between them. This organisation also maintains a direct link between the physical elements and the node hierarchy used by the post-processing application.

The final internal configuration is shown from the $+Y$ and $-Y$ sides in Figure 5.1. Both views show the distribution of the equipment across the structural plates and the different levels of the main body. The dissipative components represented in the model are the camera electronics, the battery, the On-Board Computer (OBC), the communications unit and the Attitude Determination and Control System (ADCS). Each unit is represented separately, allowing an individual internal heat load to be assigned and the corresponding temperature and heat exchanges to be analysed independently

The final external configuration is shown in Figure 5.2. The main body presents a rectangular geometry and is divided into several surface nodes. Two solar-panel wings extend from opposite sides and are discretised independently. The camera assembly extends from the main body. Heat is rejected to space through the radiator, while the MLI-covered surfaces limit heat exchange with the external environment.



(a) Internal configuration viewed from the $+Y$ side.



(b) Internal configuration viewed from the $-Y$ side.

Figure 5.1: Internal configuration of the spacecraft thermal model viewed from opposite sides. The colours correspond to the geometry layer and do not represent temperatures.

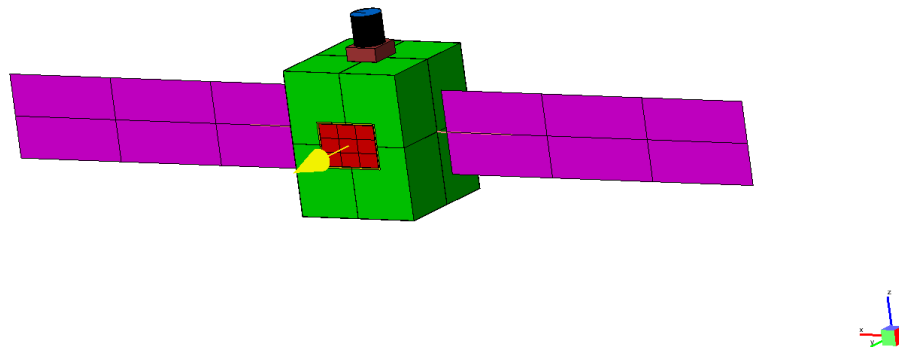


Figure 5.2: Final external configuration of the spacecraft thermal model in ESATAN-TMS. The colours correspond to the geometry layer and do not represent temperatures.

5.2. Thermal Model Hierarchy

The numerical nodes are grouped according to the physical elements they represent. This hierarchy consists of four levels:

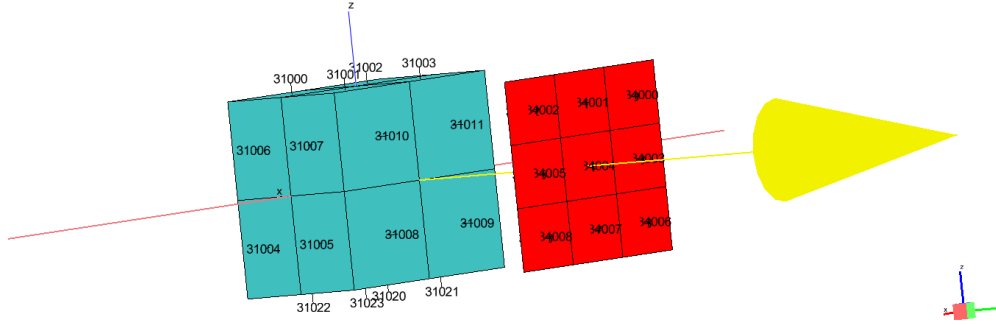
- **Level 0:** includes the complete SAT_AFM model and the SPACE boundary.
- **Level 1:** separates the main structure from the solar-panel assembly.
- **Level 2:** divides the structure into the body, camera, avionics and MLI groups.
- **Level 3:** contains the individual panels, structural plates, camera elements and equipment units.

Each group is identified by a node range and a short label, giving a consistent naming system throughout the model. The *Power* field indicates the equipment groups which have an internal heat load applied. The complete classification is given in Table 5.1.

Table 5.1: Thermal-model hierarchy and node identifiers.

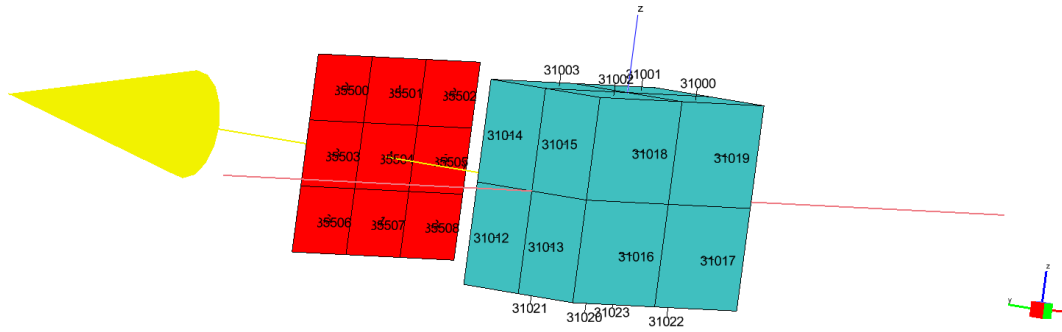
Group	Level	Parent	ID	Label	Power
Space	0	None	#99999	SPACE	False
SAT_AFM	0	None	#10000-75250	SAT_AFM	False
STRUCTURE	1	SAT_AFM	#10000-51150	STR	False
SOLAR_PANELS	1	SAT_AFM	#70000-75250	SP	False
BODY	2	STRUCTURE	#10000-13650	STR_BOD	False
CAMERA	2	STRUCTURE	#40000-43750	STR_CAM	False
AVIONICS	2	STRUCTURE	#30000-35550	STR_AV	False
MLI	2	STRUCTURE	#50000-51150	STR_MLI	False
SOLAR_PANEL_1	3	SOLAR_PANELS	#70000-72550	SP_1	False
SOLAR_PANEL_2	3	SOLAR_PANELS	#73000-75250	SP_2	False
BODY_PANEL	3	BODY	#10000-11150	STR_BOD_P	False
BODY_PLATE_INF	3	BODY	#12000-12550	STR_BOD_PLI	False
BODY_PLATE_MED	3	BODY	#12600-13050	STR_BOD_PLM	False
BODY_PLATE_SUP	3	BODY	#13100-13650	STR_BOD_PLS	False
CAMERA_BASE	3	CAMERA	#40000-40450	STR_CAM_BASE	False
CAMERA_ELEC	3	CAMERA	#40500-41050	STR_CAM_ELEC	True
CAMERA_LENS	3	CAMERA	#41500-42150	STR_CAM_LENS	False
CAMERA_CYL	3	CAMERA	#42500-43550	STR_CAM_CYL	False
BATTERY	3	AVIONICS	#30000-30550	STR_AV_BAT	True
OBC_BOX	3	AVIONICS	#31000-31550	STR_AV_OBC	True
COMMS_BOX	3	AVIONICS	#32000-32550	STR_AV_COM	True
ADCS_UNIT	3	AVIONICS	#33000-33550	STR_AV_ADCS	True
RADIATOR	3	AVIONICS	#34000-35550	STR_AV_RAD	False
MLI_PANELS	3	MLI	#50000-51150	STR_MLI_P	False

Colour, Geometry Layer



(a) First view of the OBC and radiator nodal numbering.

Colour, Geometry Layer



(b) Opposite view of the OBC and radiator nodal numbering.

Figure 5.3: Example of nodal numbering for the OBC and adjacent radiator surfaces, shown from opposite sides. The figure illustrates the node ranges listed in Table 5.1.

5.3. Material and Surface Properties

The temperature response of each geometrical element depends on the assigned bulk material and the thermo-optical properties of the surfaces. Thermal conductivity, density and specific heat capacity determine heat conduction and energy storage, while the surface properties govern the interaction with solar and infrared radiation. The same definitions are used in all analysis cases.

Consequently, changes in the calculated results are associated with the environmental and operating inputs rather than with modifications to the model.

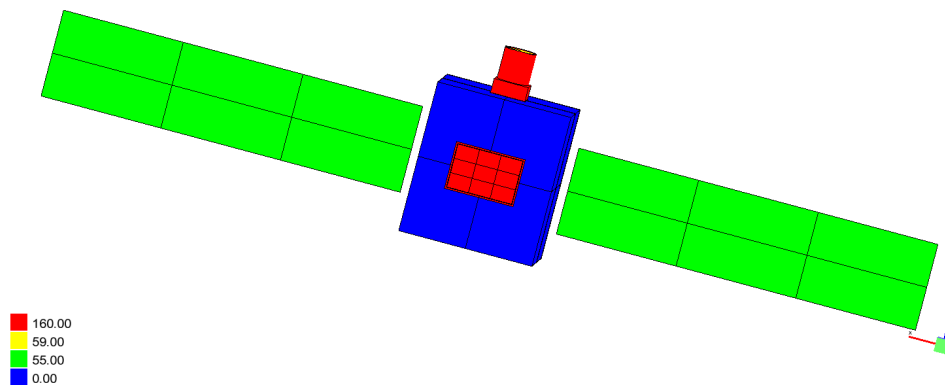
5.3.1. Bulk Properties

The geometrical elements use four material definitions. Aluminium 6061 is assigned to the main structural panels, internal plates and equipment housings. Gallium arsenide represents the solar-cell elements, while a separate optical material is used for the camera lens. The MLI foil has an independent definition whose conductivity is set to zero. This prevents the geometrical shell from introducing a solid-conduction connection, while the intended couplings are defined explicitly in the thermal network.

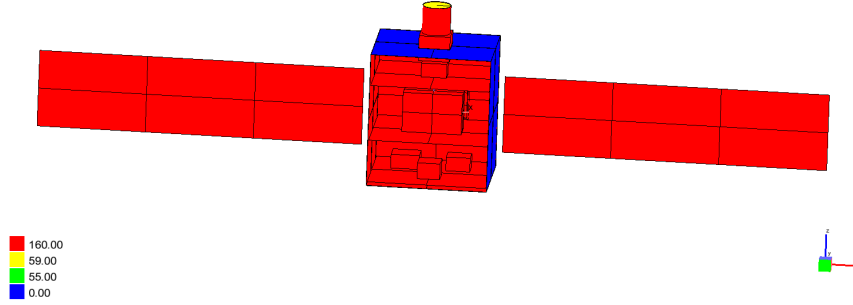
The conductivity, specific heat capacity and density assigned to each material are listed in Table 5.2. Figure 5.4 shows only the conductivity distribution, using two opposite views to cover the complete geometry. This allows the assignments to the structure, solar cells, camera and MLI surfaces to be checked.

Table 5.2: Material properties used in the thermal model.

Name	k [W m ⁻¹ K ⁻¹]	c_p [J kg ⁻¹ K ⁻¹]	ρ [kg m ⁻³]
Al_6061	160	900	2700
GaAs	55	1000	5300
Glass	59	310	5330
MLI_foil	0	900	300



(a) View from the side +Y.



(b) View from side $-Y$.

Figure 5.4: Thermal conductivity distribution over the final spacecraft model, viewed from opposite sides.

5.3.2. Thermo-optical Properties

The surface definitions are assigned according to the function of each element. Anodised aluminium is used on selected structural surfaces, while black paint is applied to the internal equipment. An Optical Solar Reflector (OSR) defines the radiator surface, and the solar cells use a separate coating definition. The external MLI is represented by Kapton coated with Indium Tin Oxide (ITO), while independent properties are assigned to the internal MLI and the infrared lens.

The solar and infrared bands are defined separately. For each surface, the model specifies absorptivity or emissivity, transmissivity and the specular and diffuse components of reflectivity. Within each band, these coefficients are defined so that the corresponding energy fractions sum to unity.

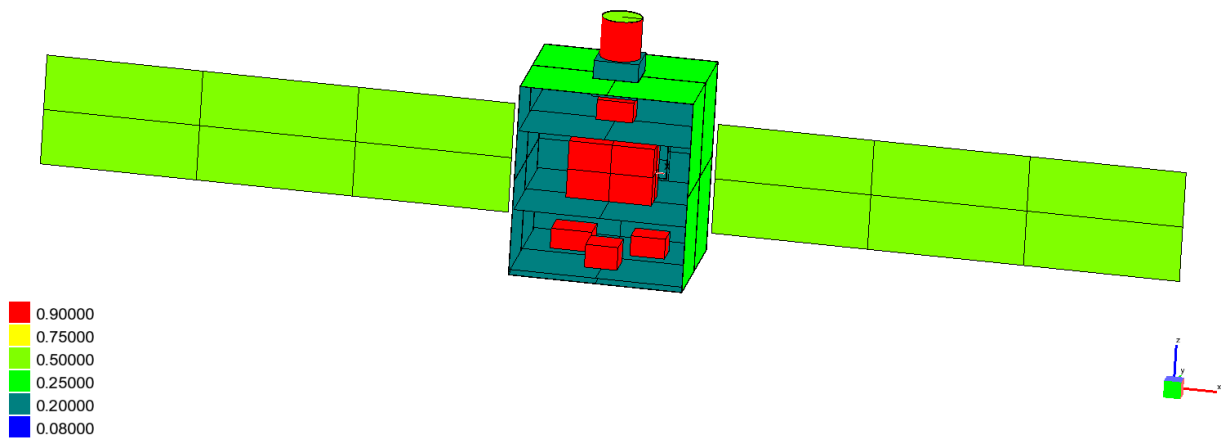
The adopted solar absorptivity and infrared emissivity values are consistent with representative measurements compiled for black coatings, anodised aluminium, solar cells, optical solar reflectors and Kapton films [17]. These data are used as reference values, since a single nominal definition is assigned to each surface in the model. The complete set of properties is summarised in Table 5.3.

Table 5.3: Thermo-optical coating properties used in the thermal model.

Name	Infrared				Solar			
	ε	τ	ρ_{spec}	ρ_{diff}	α	τ	ρ_{spec}	ρ_{diff}
Al anodized	0.6	0	0	0.4	0.2	0	0	0.8
Black paint	0.9	0	0	0.1	0.9	0	0	0.1
OSR	0.8	0	0.2	0	0.08	0	0.92	0
Solar cells	0.75	0	0	0.25	0.75	0	0	0.25
IR lens	0.1	0.9	0	0	0.5	0	0	0.5
Kapton ITO	0.8	0	0	0.2	0.5	0	0	0.5
MLI_int	0.2	0	0	0.8	0.2	0	0	0.8

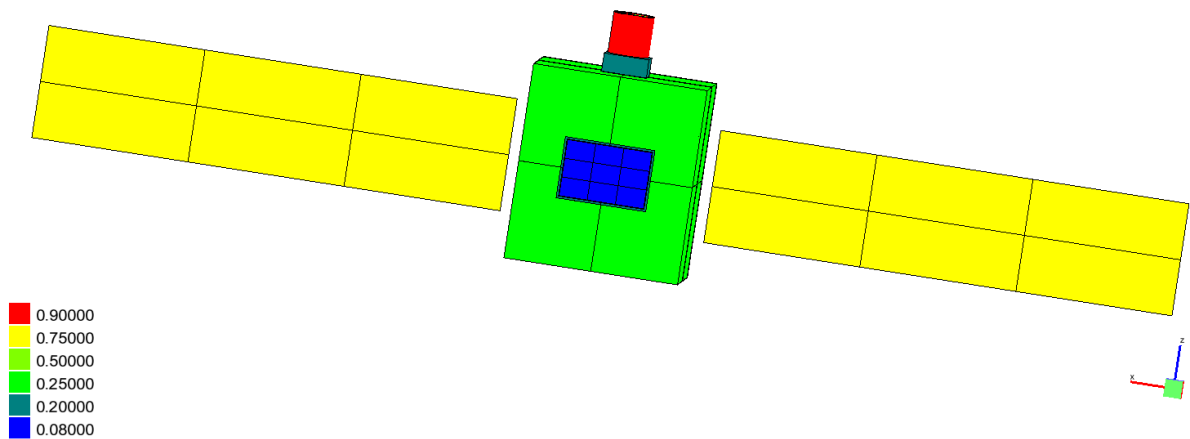
The solar absorptivity and reflectivity distributions are presented in Figures 5.5 and 5.6. Higher absorptivity is assigned to the solar cells and black-painted surfaces, whereas the radiator OSR has the lowest value. The reflectivity maps show the corresponding distribution of reflected solar energy. Two opposite views are included for each property so that the assignments can be checked over the complete geometry.

Solar Absorptivity [default], Geometry Layer



(a) View from one side.

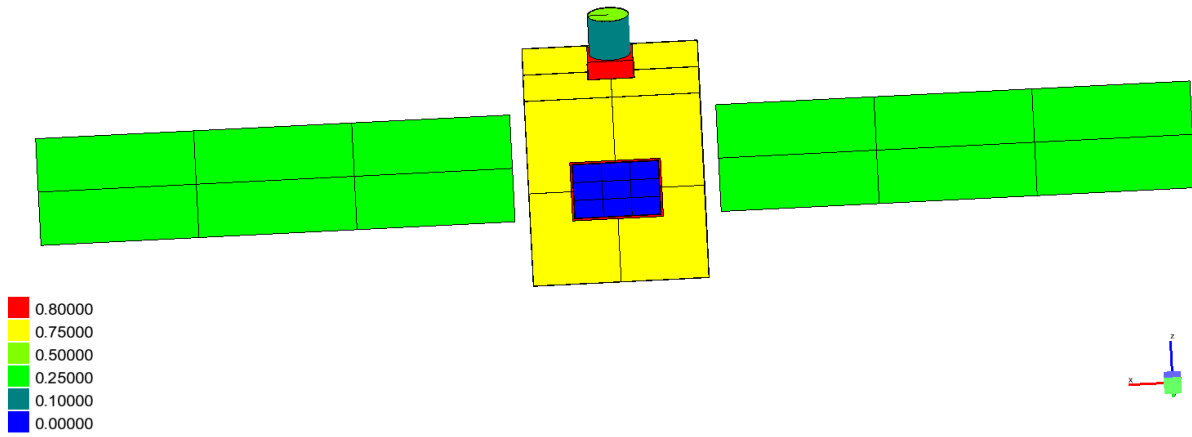
Solar Absorptivity [default], Geometry Layer



(b) View from the opposite side.

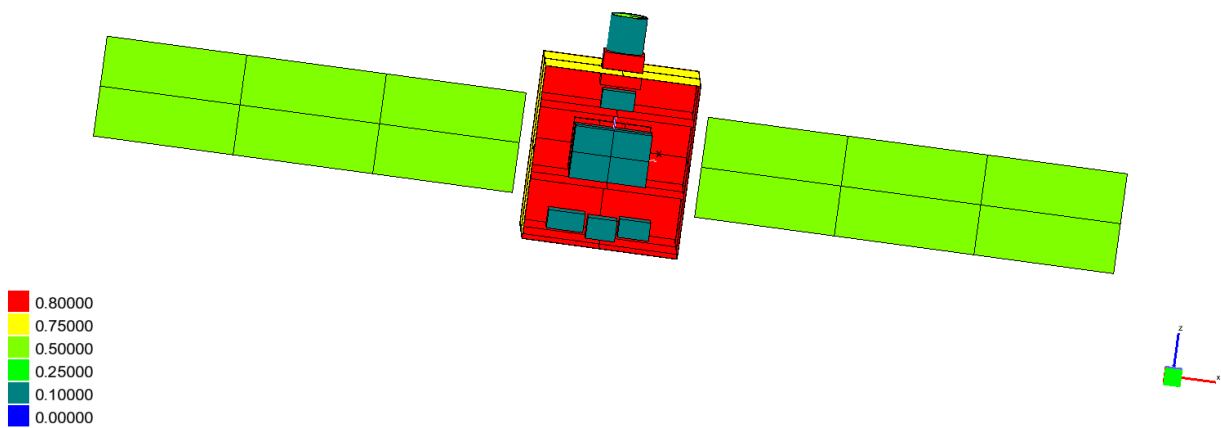
Figure 5.5: Solar absorptivity distribution over the final spacecraft model, viewed from opposite sides.

Solar Reflectivity [default], Geometry Layer



(a) View from one side.

Solar Reflectivity [default], Geometry Layer

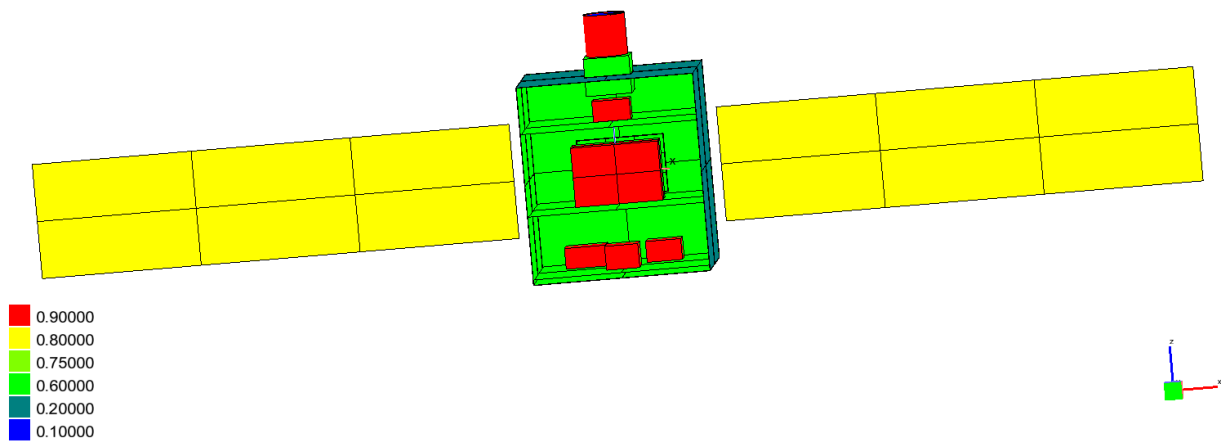


(b) View from the opposite side.

Figure 5.6: Solar reflectivity distribution over the final spacecraft model, viewed from opposite sides.

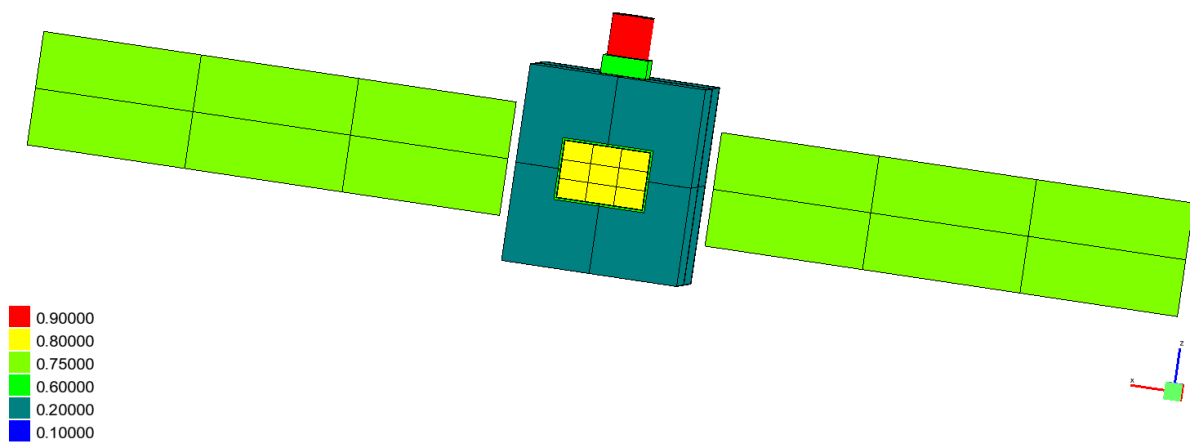
The infrared emissivity and reflectivity distributions are shown in Figures 5.7 and 5.8. High emissivity is assigned to the black-painted surfaces and the radiator, promoting radiative heat exchange. Lower values are used for the internal MLI and the camera lens, which also includes infrared transmissivity. These assignments determine the radiative exchange between internal elements and the rejection of heat from the external surfaces.

Infrared Emissivity [default], Geometry Layer



(a) View from one side.

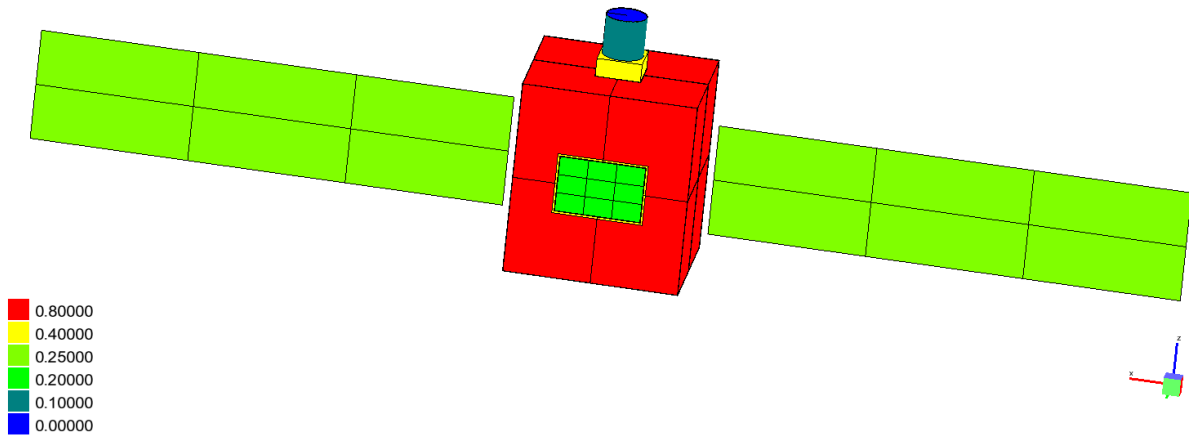
Infrared Emissivity [default], Geometry Layer



(b) View from the opposite side.

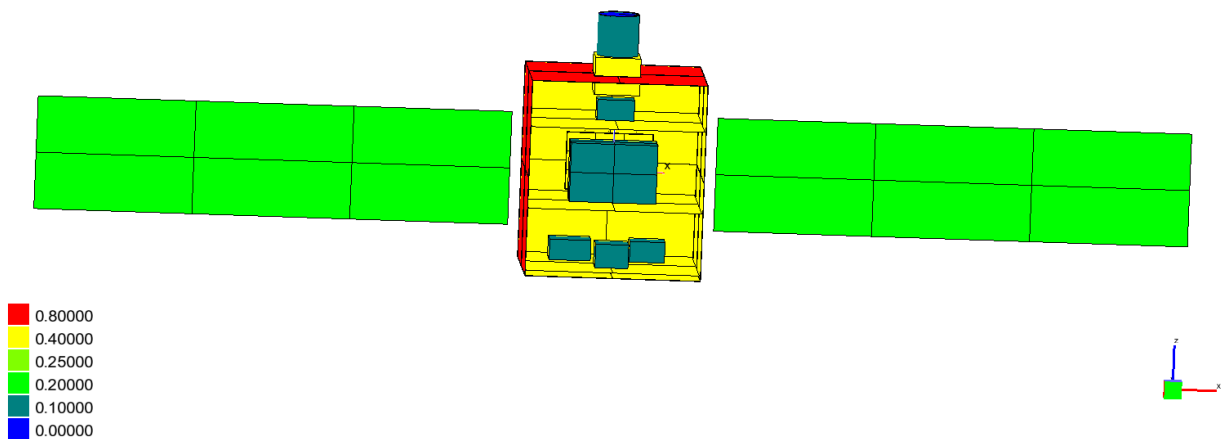
Figure 5.7: Infrared emissivity distribution over the final spacecraft model, viewed from opposite sides.

Infrared Reflectivity [default], Geometry Layer



(a) View from one side.

Infrared Reflectivity [default], Geometry Layer



(b) View from the opposite side.

Figure 5.8: Infrared reflectivity distribution over the final spacecraft model, viewed from opposite sides.

5.4. Nodal Discretisation and Thermal Network Definition

The geometrical model is divided into nodes according to the spatial resolution required for the analysis. Large external elements, including the body panels and solar-panel wings, are represented by several nodes to capture local temperature differences and variations in the absorbed environmental loads. Internal plates and equipment units are also discretised separately to reproduce the main heat-transfer paths through the structure.

Each capacitive node is assigned a material, mass and specific heat capacity. Following this nodal division, the conductive network represents heat transfer through the structure and between the internal equipment and supporting plates. Most conductive paths correspond to physical contacts included in the geometry. When a physical connection is not represented geometrically, an additional conductor is defined. This is the case for the thermal link between the OBC and the radiator, which is included without adding unnecessary geometrical detail.

The radiative network represents exchange between the relevant surfaces and with the deep-space boundary.

Internal heat loads are applied to the nodes representing the dissipative equipment. The values assigned in each transient case are summarised in Table 5.4. The battery load is defined separately for eclipse and non-eclipse intervals.

Table 5.4: Internal heat loads applied in the selected transient cases.

Component	Applied power (W)		
	Nominal	Hot	Cold
Camera electronics	2.00	3.50	0.00
Battery, outside eclipse	1.50	2.06	0.375
Battery, during eclipse	10.00	13.75	2.50
OBC	4.00	5.00	2.00
Communications unit	3.00	4.50	0.50
ADCS	2.00	3.50	0.50

The node definitions and conductive and radiative couplings remain unchanged in all transient cases. Only the environmental conditions and internal heat loads are modified as requires for each case. Consequently, each node and thermal coupling represents the same physical element in every exported result file, allowing direct comparison between cases.

5.5. Orbital Environment and Radiative Cases

The orbital environment defines the external radiative conditions applied to the model. Direct solar radiation, planetary albedo and planetary infrared emission are included as external heat inputs [18]. Energy emitted by the external surfaces towards deep space is also considered. These exchanges are

represented through the orbital and radiative models of ESATAN-TMS and provide the boundary conditions used in the transient simulations [8].

Three radiative cases are selected: `rad_nominal`, `rad_hot_operational` and `rad_cold_operational`. The same geometry and surface properties are retained in all three cases, while the environmental parameters and eclipse duration differ between them. For the radiative calculation, each orbital revolution is divided into 24 positions separated by 15° . This common angular sampling allows the incident loads to be evaluated consistently throughout the orbit and included in the transient solution.

Figure 5.9 shows the orbital configuration of the nominal case. The spacecraft positions illustrate the discretisation used for the radiative calculation. The green segment represents the portion outside eclipse, whereas the red segment identifies the eclipse interval.

Colour, Geometry Layer

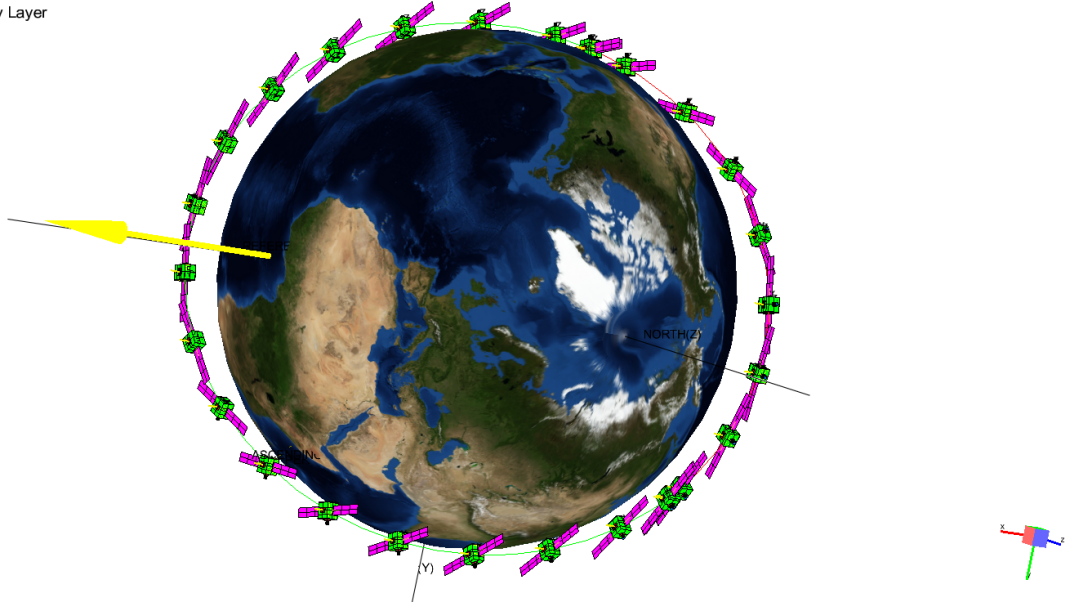


Figure 5.9: Nominal orbital configuration and the 24 positions used for the radiative calculation. The green segment represents the portion outside eclipse, while the red segment indicates the eclipse interval.

The `rad_nominal` case provides the reference environment. `rad_hot_operational` combines a higher external heat input with a shorter eclipse, whereas `rad_cold_operational` combines a lower input with a longer eclipse.

These environments provide clearly differentiated conditions for comparing the hot and cold responses with the nominal case without modifying the physical model. They are selected for the analyses performed in this work and do not cover every condition expected during a flight mission.

5.6. Transient Analysis Cases

5.6.1. Definition of Analysis Cases

Three transient simulations are selected for the final analysis: nominal, hot and cold. The nominal case provides the reference response, while the hot and cold cases represent higher and lower external heat inputs, respectively. Different equipment loads are also assigned according to the operating conditions defined for each simulation. Their purpose is summarised in Table 5.5.

Table 5.5: Transient cases selected from the common thermal model.

Case	Radiative case	Purpose
Nominal	rad_nominal	Reference response used for comparison with the hot and cold cases.
Hot	rad_hot_operational	Response under higher external heat input and a shorter eclipse.
Cold	rad_cold_operational	Response under lower external heat input and a longer eclipse.

The same geometry, materials, surface properties, node labels, hierarchy and thermal couplings are retained in all three simulations. Only the radiative environment and internal heat loads change between cases. Equivalent nodes and variables therefore retain the same physical meaning, allowing the hot and cold results to be compared directly with the nominal case.

5.6.2. Solver Configuration and Temporal Sampling

The *Solution Control* configuration applied to the three analysis cases is shown in Figure 5.10.

```

C
    TIMEND=6050.2473
    OUTINT=60.502473
    NLOOP=100
    RELXCA=0.01
    DTIMEI=1.0
    CALL SOLCYC('SLCRNC',0.01D0,0.1D0,6050.2473D0,1000,' ','NONE')
C
    CALL SLCRNC
C

```

Figure 5.10: *Solution Control* execution block used to obtain cyclic convergence and calculate the final transient orbital period in ESATAN-TMS.

The same numerical settings are used in all three simulations. The transient solution is calculated using the Crank–Nicolson method. Before the final results are exported, SOLCYC repeats the orbital period until the maximum difference between the initial and final nodal temperatures is below $0.01\text{ }^{\circ}\text{C}$. A maximum of 1000 cycles is allowed. The converged temperature field is then used as the initial condition for the orbital period retained for analysis.

Each exported simulation covers 6050.247 s and contains 101 output instants separated by approximately 60.50 s. This common time basis records the main changes in the applied loads and allows values from different cases to be compared at equivalent points of the orbit.

5.7. Thermal Model Verification

Verification is performed throughout model development using visual and numerical checks. The material and surface-property maps are inspected to identify missing or incorrectly assigned faces. The node hierarchy is compared with the physical arrangement to confirm that the external panels, internal plates and equipment units belong to the correct model groups.

The conductive and radiative networks are reviewed to identify isolated nodes, missing links and unintended heat-transfer paths. Nodal energy balances are examined to detect inconsistent loads or exchanges. Finally, the exported results are checked to confirm that every selected case satisfies the required cyclic-convergence criterion.

These checks confirm the internal consistency of the model but do not validate it against experimental data. No corresponding flight hardware or thermal-vacuum test data are available for comparison. The conclusions are therefore limited to the behaviour predicted by the model under the defined environmental and operating conditions.

5.8. Simulation Results Export in TMD Format

After each simulation is completed, the result set obtained for each case is exported from ESATAN-TMS as a separate .TMD file. Each file stores the time vector and the nodal quantities selected for post-processing, including temperatures, applied loads, absorbed environmental loads, and conductive and radiative heat flows. Node numbers and labels are preserved during export, allowing each variable to be associated with the corresponding group in the external hierarchy spreadsheet.

The structural and temporal dimensions shared by the three result files are summarised in Table 5.6. The nominal, hot and cold files retain the same node and conductor structure and cover a common time interval.

Together, the three files form the dataset used for the transient and multi-case analyses. Their common structure allows equivalent variables to be represented at the same orbital instants and compared directly with respect to the nominal case.

Table 5.6: Structural and temporal characteristics of the exported .TMD files.

Total thermal nodes per file	284
Capacitive nodes	282
Boundary nodes	1
Inactive nodes	1
Linear conductors, G^L	597
Radiative conductors, G^R	3292
Time interval	6050.247 s
Output instants per file	101
Approximate output time step	60.502 s

The 101 output instants correspond to the temporal sampling of the transient solution and should not be confused with the 24 orbital positions used in the radiative calculation. For each nodal variable, one file therefore contains $284 \times 101 = 28\,684$ values before the conductor results and remaining nodal quantities are considered.

The absorbed solar loads calculated for the solar panels are also retained in the exported data. These values are used as inputs for the electrical-generation and battery calculations. This processing is performed independently and does not modify the ESATAN-TMS solution.

The three .TMD files and the hierarchy spreadsheet form the input dataset used by the Python application to read, organise and compare the results.

6. Development of the Multi-Case Post-Processing Tool

The original post-processing application is extended to handle several transient cases associated with a single ESATAN-TMS model. The hierarchy-based selection and heat-flow representation functions already available are adapted to analyse selected transient instants, while the workflow is expanded to support multi-case loading, comparison and reusable outputs.

Each .TMD file is treated as an independent case and linked to the model hierarchy defined in a common Excel spreadsheet for all cases. This arrangement allows equivalent elements to be examined using consistent selection, time and representation criteria across all loaded simulations.

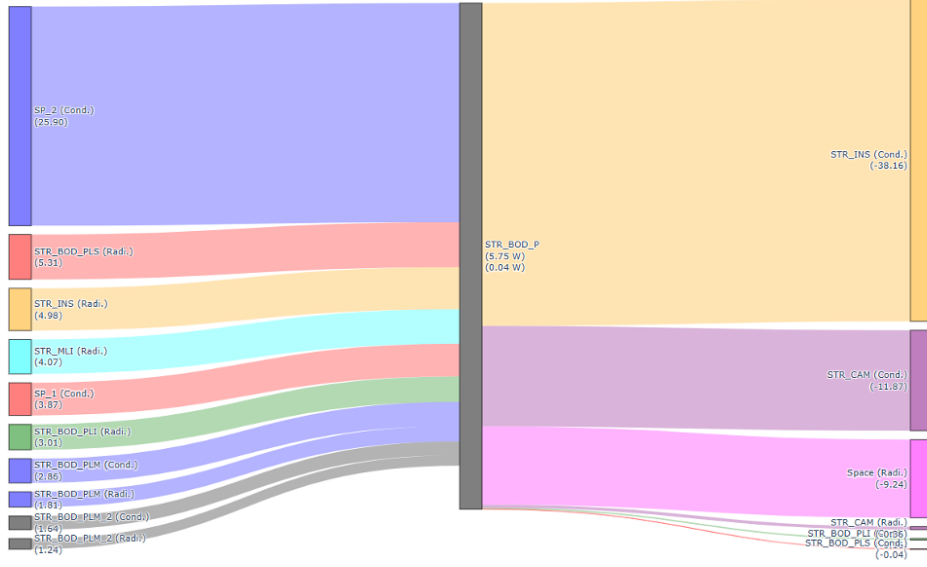
The nodal model is solved exclusively in ESATAN-TMS. The post-processing tool neither repeats the thermal calculation nor modifies the imported data. Its purpose is to organise these values and provide the functional controls required for inspection, transient post-processing, multi-case comparison and electrical assessment. The capabilities specific to the transient and energy-balance modules are developed in the following chapters.

6.1. Evolution of the Original Post-Processing Application

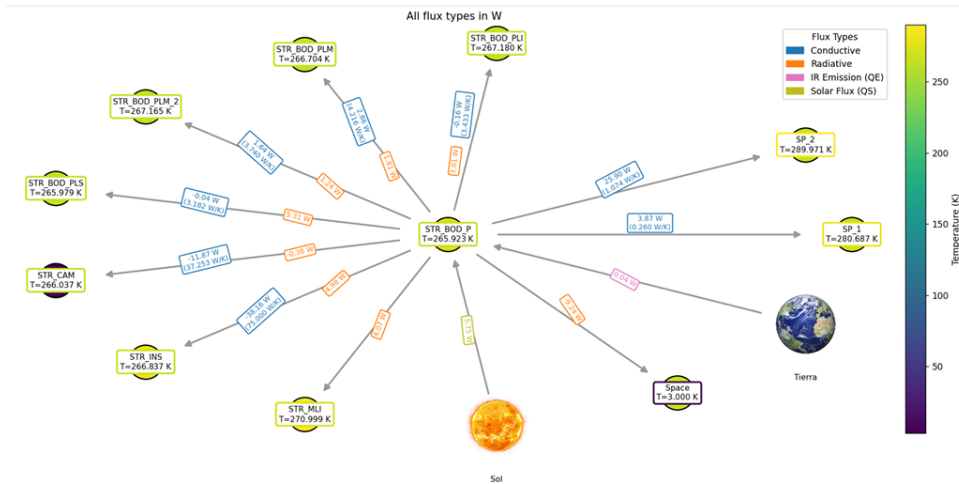
The starting point is the Python-based code developed for the steady-state post-processing of ESATAN-TMS models. Its workflow combines the data contained in a .TMD file with an Excel spreadsheet that associates node identifiers with physical components and subsystems. This mapping supports result filtering and the generation of temperature views, group-attribute and conductor tables, and heat-flow representations.

The result reader, hierarchical selection routines and main heat-flow views form the functional basis for the present development. These capabilities can also be applied to transient inspection, since the state of a solved model can be examined at a chosen output instant. Figure 6.1 illustrates two representative visualisations generated for the HELIOS steady-state case [1].

STR_BOD_P Heat Balance [W](tol=1.0e-03)
(54.67 W) + (5.79 W) = (60.46 W)
(-59.82 W)



(a) Sankey representation of the HELIOS heat balance.



(b) NX diagram of HELIOS including all heat-flow types.

Figure 6.1: Representative heat-flow visualisations generated for the HELIOS steady-state case with the original application. Extracted from [1].

The original interface is mainly designed for the inspection of a single result file. Although some routines already allow access to several output instants when they are present in the source file, most of the interface is oriented towards steady-state analysis. In addition, saved figures and tables do not record the active case, involved elements or output instant used to re-create them, so the associated configuration cannot be restored directly.

The current implementation incorporates several new functions that expand the capabilities of the

application. Multiple compatible cases remain accessible within a single session, equivalent elements are selected through a shared hierarchy, and instantaneous outputs are generated at a single selected transient time. Saved figures and tables also store the case, element and representation settings required to restore the saved configuration.

Accordingly, the existing reading, filtering and representation functions are integrated into an expanded workflow. The resulting tool accepts multiple `.TMD` files associated with a shared model structure and incorporates the transient and electrical-balance modules required to meet the objectives of the project. Nominal, hot and cold simulations serve as representative cases, but the implemented workflow is not limited to these specific conditions.

6.2. Application Architecture and Data Flow

The software follows a modular organisation that separates data interpretation, interface coordination and the individual functional areas. The architecture is described through the path followed by the imported data rather than as a detailed inventory of internal storage or execution mechanisms.

Execution starts in `app_thermalviewer/app_interactiva.py`, which receives the shared hierarchy, the internal analysis identifier and the selected `.TMD` files. This entry point builds the collection of loaded cases and distributes it among the tabs forming the user interface.

`core/thermaldata.py` acts as the data layer. It interprets each result file and gives the interface access to nodal, group, time-dependent and conductor information. Individual `.TMD` files are processed separately, while the Excel mapping defines the component organisation required to select equivalent elements.

Shared operations are grouped in `app_helpers.py` and `viewer_utils.py`. These modules handle hierarchy ordering, selection validation, unit management, time-dependent access, and direct or difference-based representations across several cases.

The files contained in `tabs/` apply these operations to the controls, tables and figures associated with each functional area. Each file implements one of the main interface tabs. For example, `tab_case_info.py` displays general information about the loaded simulations, `tab_flow_explorer.py` handles heat-flow inspection, `tab_transient.py` implements the transient post-processing module, and `tab_sp.py` contains the electrical-balance and battery calculations. Further modules, including `tab_group_attributes.py`, `tab_gl_conductors.py`, `tab_gr_conductors.py`, `tab_nodes.py` and `tab_group_study.py`, cover the remaining model-inspection and conductor-analysis areas.

A final group of modules handles the saving, restoration and export of analyses throughout the interface. Each generated figure or table is linked to the main settings needed to reproduce or export it. Table 6.1 summarises the functional responsibilities of the principal software groups.

Table 6.1: Main software groups and their functional responsibilities.

Module or directory	Main responsibility
<code>app_interactiva.py</code>	Coordinates file inputs, loaded cases and execution of the analysis tabs.
<code>core/thermaldata.py</code>	Reads each result file and provides access to nodes, attributes, time histories and conductor data.
<code>app_helpers.py</code> , <code>viewer_utils.py</code>	Provide hierarchy management, selection validation, unit handling, case-comparison operations and common output preparation.
<code>tabs/</code>	Contains one module for each analysis tab, including case inspection, heat-flow exploration, conductor analysis, transient post-processing, and the electrical balance and battery model.
<code>scene_manager.py</code> , <code>scene_ui.py</code> , <code>restored_scenes.py</code>	Provide the controls required to save, list, restore and export complete analysis configurations and their generated outputs.

Figure 6.2 outlines the resulting data flow. The input files are interpreted separately, coordinated through a central application layer and passed to the relevant processing modules. This structure links the generated figures and tables to the saving and export controls available in each area.

Software architecture of the final post-processing application

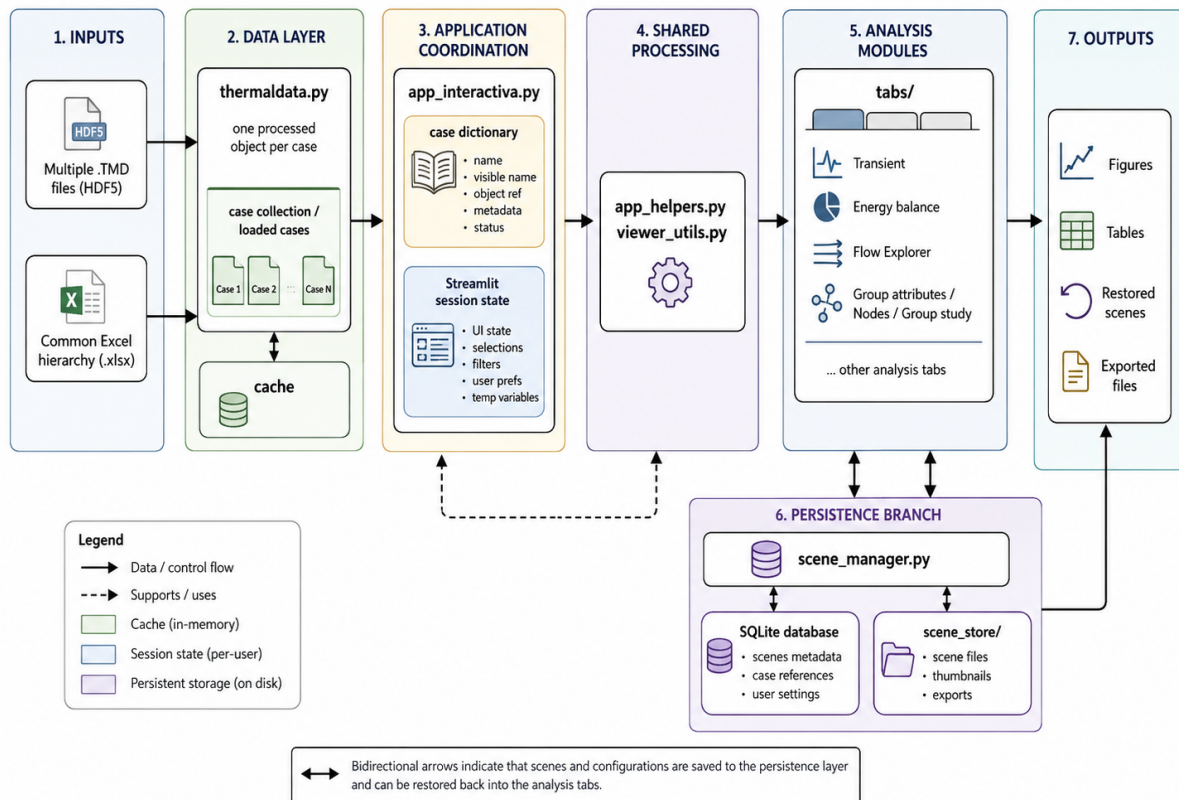


Figure 6.2: Main data flow and software architecture of the current post-processing application.

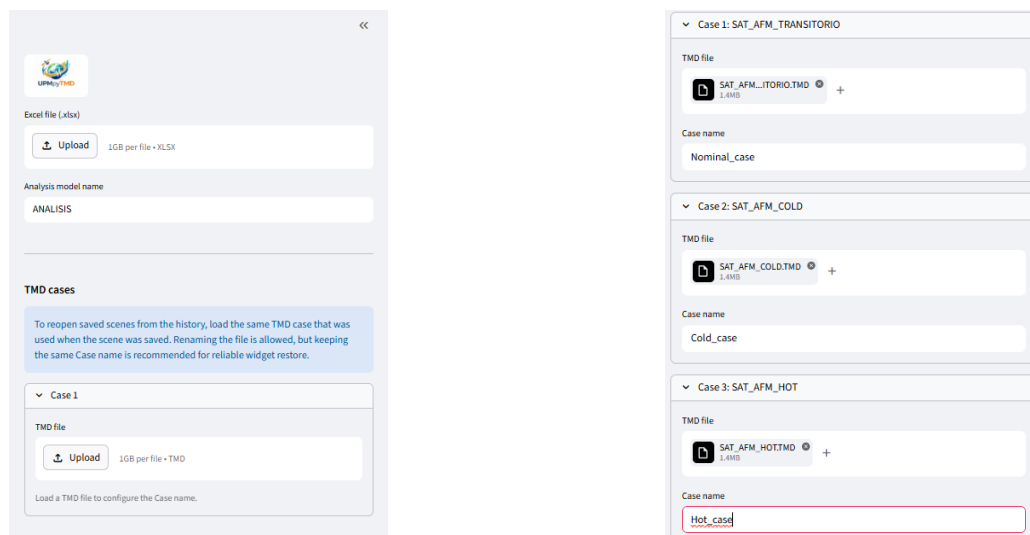
Imported simulation data are kept separate from the values calculated during post-processing. Changes to the active groups, cases, units, output time or representation mode affect only the current analysis and leave the imported .TMD files unchanged. Consequently, every tab can access the loaded case collection and apply consistent selections without creating modified copies of the source data.

6.3. Multi-Case Loading and Case Organisation

The loading stage combines a single Excel hierarchy with multiple transient .TMD files. One spreadsheet serves all simulations because they share identical node identifiers, physical groups and parent-child relationships. Environmental and operating conditions change between simulations, but the model organisation remains unchanged.

Each .TMD file contains an internal analysis identifier, while the interface assigns a visible name to the loaded simulation. Different files may therefore use an identical internal identifier, such as ANALISIS, but appear as *Nominal*, *Hot* or *Cold* in selectors, legends, tables and saved analyses.

Before loading a file, the interface provides an editable field for its visible name. If two entries receive identical names, a suffix is added to keep every key in the case collection unique. Figure 6.3 shows the initial inputs and the configuration of multiple result files.



(a) Initial hierarchy and result-file inputs.

(b) Configuration of several files through editable case names.

Figure 6.3: Multi-case loading interface with one shared hierarchy and multiple .TMD result files.

The loader processes every file separately and stores it as an independent case. Any compatible tab can then request the corresponding variable or component selection from one or more simulations without combining or modifying their original numerical data.

The Excel hierarchy does more than replace node identifiers with readable names. It maps each physical group to its node range and defines its position within the parent–child structure of the model. Since all loaded simulations use this mapping, one component selection can be applied directly to each of them.

For example, after the user selects **BODY**, the tool resolves its node range once. Each simulation then supplies the values of those nodes at the requested time, without requiring the numerical identifiers to be entered again for the nominal, hot and cold simulations. The resulting tables and curves therefore refer to the corresponding region before any overlay or difference calculation is performed.

The hierarchy also controls ordering and prevents double counting. Selectors follow the hierarchical parent–child sequence defined in Excel rather than a purely alphabetical list. When a parent group and one of its descendants are chosen together, the application removes the descendant because the parent already contains its nodes. A visible warning identifies the corrected selection. This check is particularly important for heat-flow and aggregated load results, where repeated nodes would otherwise alter the numerical total.

6.4. Functional Adaptation of the Application Interface

The final version brings the inherited inspection tabs and the newly developed modules together within a single interface. The sidebar contains the inputs required throughout the session, while every tab receives the loaded case collection and shared Excel hierarchy. The user can therefore move between model inspection, transient and electrical post-processing without loading the source files again or redefining the model structure.

Figure 6.4 presents the resulting layout after loading the Excel hierarchy spreadsheet and three transient cases. The sidebar groups the model inputs, whereas the main area provides access to the different tabs and their case, element, time and representation controls.

The existing group of tabs covers model inspection and the heat-flow functions offered by the original tool. **Case info** summarises the datasets, groups and couplings contained in the loaded simulations. **Group attributes**, **Nodes** and **Group study** provide different levels of detail for the chosen parts of the model. **GL conductors** and **GR conductors** examine conductive and radiative connections, respectively. **Flow Explorer: Multilevel Comparison** uses equivalent component selections to generate instantaneous Sankey and network representations.

Most changes introduced in these tabs affect their operation rather than their appearance. Each compatible module now reads from the loaded case collection, applies a consistent component selection and, when transient data are available, evaluates the requested information at a chosen output instant.

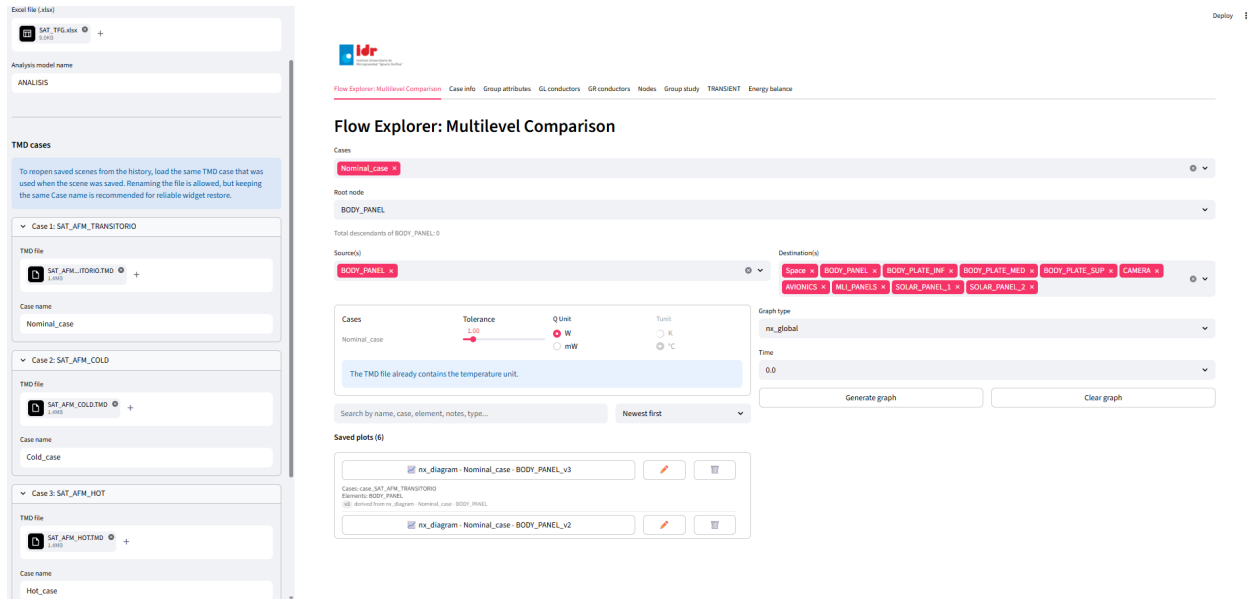


Figure 6.4: Final post-processing interface after loading three transient cases. The sidebar groups the shared inputs, while the main area provides access to the model-inspection, transient and electrical-analysis modules.

Enlarged views of these functional areas are included in Appendix A. Each tab is presented separately to provide a readable view of its case, hierarchy, time and representation controls.

The controls shown by each tab depend on the information present in the active case. Temperature units already defined in the result file remain fixed, whereas compatible heat-flow and display units can be changed when required. Before generating a figure or table, the tool checks for empty case selections, unsupported combinations and missing data. Clear messages identify the problem and prevent the calculation from continuing with incomplete inputs. These checks give all compatible modules a consistent operating logic.

6.4.1. Time Selection

The inherited tabs represent the state of the model at one output instant, even when the source file contains a complete transient solution. When multiple output steps are present, each compatible tab adds a time selector. The module reads the stored time vector and applies the chosen instant to group attributes, nodal tables, conductor summaries and instantaneous heat-flow representations.

This adaptation extends the original steady-state views without changing their interpretation. Instead of replacing them with complete time histories, the user can inspect any stored output instant with the defined component selection and graphical tools. Figure 6.5 illustrates the selector at the beginning and end of a stored time vector.

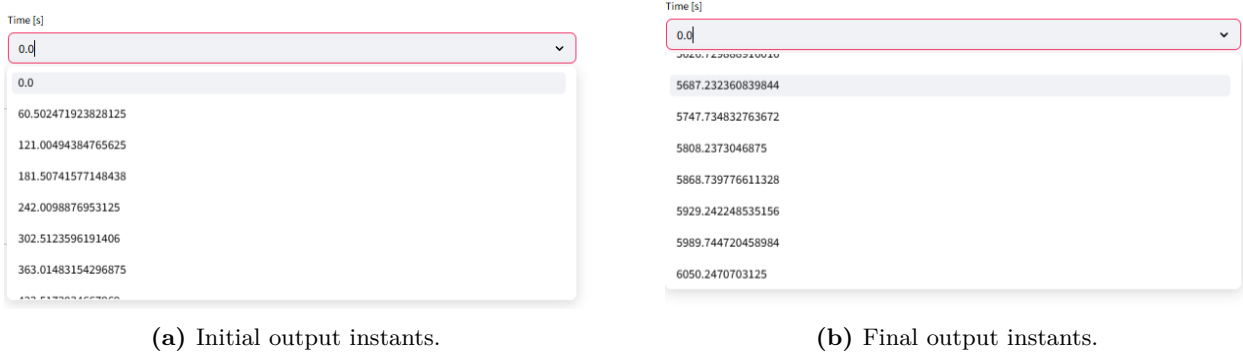


Figure 6.5: Output-time selector used by the instantaneous inspection modules when transient data are available.

When multiple simulations are active, the module applies the chosen instant to each compatible simulation. Saving the analysis also records this setting, so restoring it returns the time control to its previous position.

6.4.2. Multi-Case Representation Options

The case selector allows each compatible tab to work with one or more loaded simulations. In direct representation mode, the tool applies consistent selection and temporal settings to every chosen case. Curves can be superimposed when a shared graph provides a clear comparison, whereas tables and complex network representations remain separated when combining them would reduce readability. The additional *Case difference* mode isolates the numerical change between simulations. The user chooses one reference case and one or more comparison cases. The tool checks that the requested variables are compatible and generates one difference result for each reference–comparison pair. Figure 6.6 presents the representation mode and the associated case controls.

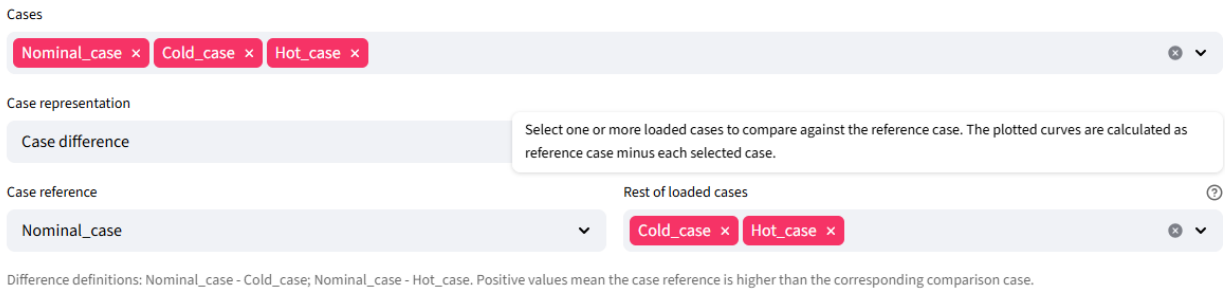


Figure 6.6: Configuration of the case-difference representation.

For any compatible quantity x , the adopted convention is

$$\Delta x_{R-C} = x_R - x_C, \quad (6.1)$$

where R denotes the reference case and C one comparison case. A positive value indicates that the reference result is greater, while a negative value indicates that the comparison result is greater. The reference-comparison order is used consistently in legends, table headings, captions and saved analyses.

Direct representation and case difference serve different purposes. An overlay keeps the absolute response of each simulation visible and makes shared trends easier to identify. The difference removes the common level and highlights the effect of a change in environmental or operating conditions. Both modes use the defined component selections, units and time settings, so the user can change the representation without entering the complete configuration again.

The last two tabs contain the main modules introduced in this work. **TRANSIENT** processes complete histories of temperatures, loads, interfaces and heat flows. **Energy balance** calculates solar generation, electrical demand and battery behaviour from the chosen simulations. The following chapters develop the calculations and outputs of both modules separately.

6.5. Saving, Restoring and Exporting Analyses

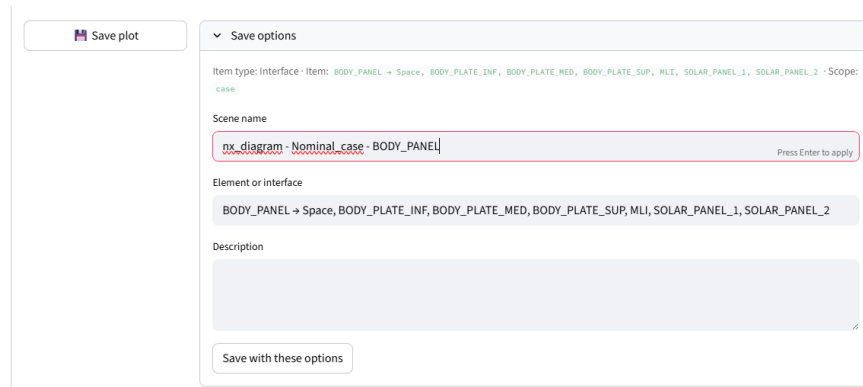
Combining several simulations, involved elements, variables, time instants and representation settings can produce a large number of valid outputs. Exporting a figure or table preserves the final result, but not the complete configuration needed to reproduce it inside the interface. For this reason, the compatible tabs include an area for saving and restoring their generated outputs.

Once a figure or table has been created, the corresponding saving control becomes active. Depending on the result type, the option appears as *Save plot*, *Save table* or as a combined control when both outputs belong to a single analysis. The user assigns a descriptive name and may add a short explanation before saving it.

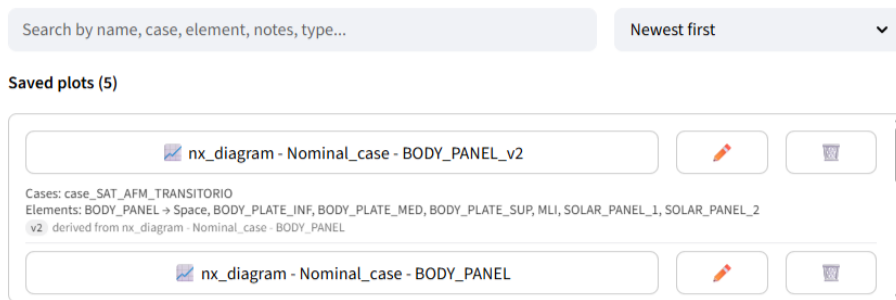
Each saved analysis, referred to as a scene within the interface, records the source tab together with the cases, involved elements, variables, units, time instant and representation mode used to produce the result. These settings allow the stored configuration to be recovered without repeating the complete selection process.

Saved analyses use two coordinated storage levels. The SQLite database `scene_memory.db` records the model, cases, source tab, configuration settings and paths associated with each entry, while the local `scene_store/` directory contains the generated files. Each scene stores its metadata and configuration in `scene.json`; Plotly figures are saved in JSON and HTML formats, Matplotlib figures in PNG and SVG, and tables in CSV and XLSX. The database entry and its associated local folder must remain consistent for the complete configuration to be restored.

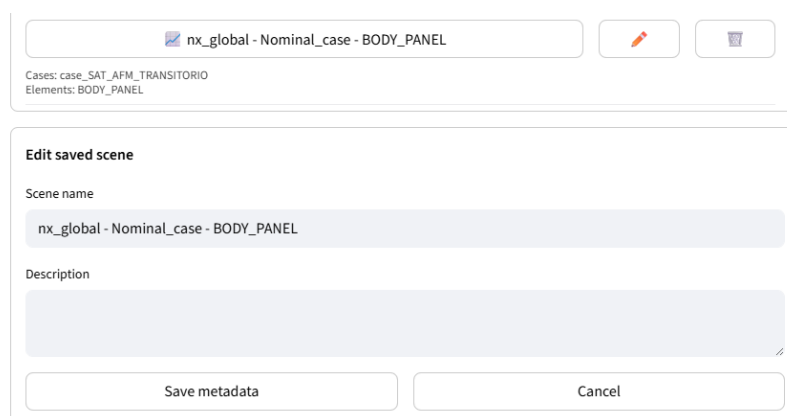
Figure 6.7 illustrates the main operations available after generating an output. The first view contains the information entered before saving it. The second shows an original analysis and a later version derived from it, while the third illustrates the editing of its identifying information.



(a) Definition of the scene name, involved elements and optional description before saving the output.



(b) Saved-analysis list containing an original result and a subsequently derived version. Selecting the displayed name restores its configuration, while the adjacent controls allow the entry to be edited or deleted.



(c) Editing of the name and description associated with an existing scene.

Figure 6.7: Sequence used to save, restore and manage a generated output.

Selecting the name of a stored entry returns its main settings to the tab from which it was created. The source files and required cases must already be loaded, but the physical selections, variables,

units, output time and representation options are recovered automatically. The controls placed next to each entry allow its name and description to be modified or the complete entry to be removed.

Restoring an analysis does not overwrite the saved result. If the recovered configuration is modified and stored again, the tool creates a new entry and identifies it as a derived version. The previous output remains accessible, while the new one can use another case, time instant, variable or representation setting.

Search and ordering controls simplify the management of larger collections. Saved analyses can be located through their name, case, model element, description or output type. Although all compatible tabs follow a consistent saving procedure, each area displays the entries and files associated with its own calculations.

Figures and tables can also be exported using the formats offered by their source module. These files provide conventional outputs for reports, engineering documentation or later processing outside the interface.

Scene storage and file export fulfil complementary purposes. The saved-analysis system preserves the configuration and relationships required to continue the work inside the application. Download controls create external copies of figures or tables for reports and engineering documentation. Together, both functions connect interactive exploration with reproducible post-processing and final result presentation.

The resulting chapter-level workflow can be summarised as follows: the application loads multiple compatible cases, associates them with one hierarchy, applies consistent selections and time settings, generates direct or difference outputs, and stores the complete configuration whenever it must be recovered. This architecture provides the shared foundation for the transient and electrical modules developed next.

7. Transient Thermal Analysis

The multi-case structure developed in Chapter 6 provides the model hierarchy, case collection, time controls and representation options used by the analysis modules. Within this framework, the **TRANSIENT** tab processes the time-dependent temperatures, thermal loads and conductor data stored in the imported ESATAN-TMS .TMD files. The original simulation results remain unchanged, and all additional calculations are performed during post-processing.

The functions available in the tab are described using the nominal case `SAT_AFM_TRANSITORIO.TMD`. The analysis covers temperature and thermal-load evolution, interface temperature differences, variations over selected time windows, summary values, conductive and radiative heat flows, and accumulated thermal energy. Comparisons between different environmental and operating cases are reserved for the results chapter.

7.1. Temperature and Thermal-Load Evolution

The first area provides a general view of the transient response through the evolution of temperatures, thermal loads and selected group attributes. To build these time-dependent representations, the application uses the Excel hierarchy to associate each physical group g with its corresponding set of nodes $\mathcal{N}_g$.

At every output instant, the minimum, mean and maximum temperatures of the group are calculated as:

$$T_{g,\min}(t) = \min[T_n(t)]_{n \in \mathcal{N}_g} \quad (7.1)$$

$$T_{g,\text{mean}}(t) = \frac{1}{|\mathcal{N}_g|} \sum_{n \in \mathcal{N}_g} T_n(t) \quad (7.2)$$

$$T_{g,\max}(t) = \max[T_n(t)]_{n \in \mathcal{N}_g} \quad (7.3)$$

Thermal loads are additive parameters, so they are combined through a different aggregation rule. For any available contribution Q_x , the corresponding group evolution is obtained from:

$$Q_{x,g}(t) = \sum_{n \in \mathcal{N}_g} Q_{x,n}(t), \quad x \in \{A, E, S, I, R\}. \quad (7.4)$$

Here, Q_A , Q_E and Q_S denote the absorbed albedo, planetary infrared and direct-solar loads, respectively, while Q_I and Q_R represent the internal and additional applied loads exported with the solved model. Together with the temperature parameters defined above, these aggregated series set the basis for the subsequent analyses available in the tab.

The configuration begins with the selection of one or more cases and model elements. For each element, the user then chooses the parameters to be represented. The available options are $T_{\text{máx}}$, T_{mean} , $T_{\text{mín}}$, Q_A , Q_E , Q_S , Q_I , Q_R , C and A . The three temperature parameters are activated by default, although the user can select any other combination.

The temperature parameters describe the thermal state of the selected element. T_{mean} represents its overall evolution, while $T_{\text{máx}}$ and $T_{\text{mín}}$ identify the highest and lowest nodal temperatures at each output instant. The distance between both limits defines the instantaneous temperature range within the corresponding group.

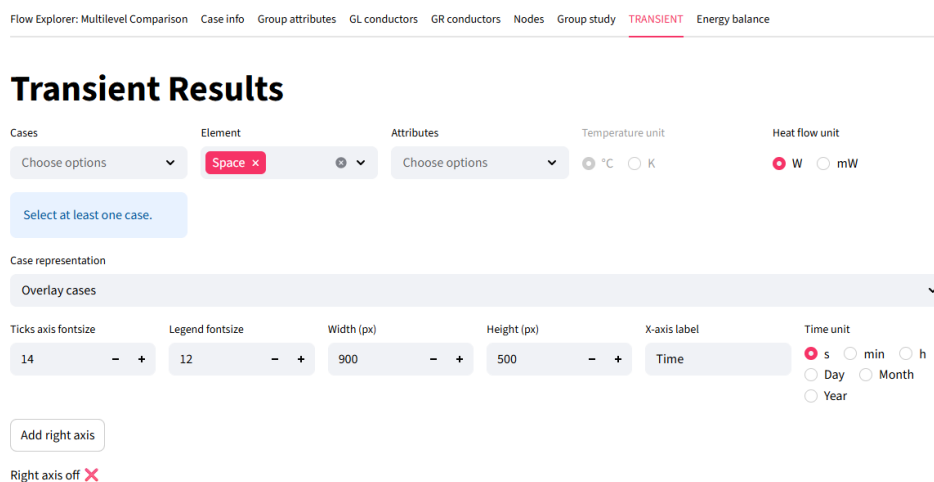
The thermal-load parameters can be represented separately or together with the temperature parameters. Their curves show how the environmental and internally applied loads acting on the selected element vary throughout the simulation. When an element contains several nodes, the displayed value corresponds to the group sum defined in Equation (7.4).

Thermal capacity C and surface area A can also be displayed. Unlike the temperatures and loads, these properties remain constant throughout the transient simulation and are therefore not used in the time-dependent examples presented below.

The display units are configured independently from the parameter selection. Temperatures are restricted to degrees Celsius, while thermal loads can be represented in watts or milliwatts, and elapsed time in seconds, minutes, hours, days, months or years.

Further controls modify the horizontal-axis label, graph width and height, axis tick font size and legend font size. The interface also provides the multi-case representation alternative introduced in Section 6.4.2. The examples presented here use a single case, since this is sufficient to explain the functionality of this area of the tab.

Figure 7.1 shows the complete configuration area used to define the cases, elements, parameters, units and graphical settings of the resulting plot.



Flow Explorer: Multilevel Comparison Case info Group attributes GL conductors GR conductors Nodes Group study **TRANSIENT** Energy balance

Transient Results

Cases: Choose options

Element: Space

Attributes: Choose options

Temperature unit: ☒ °C ☐ K

Heat flow unit: ☒ W ☐ mW

Case representation: Overlay cases

Ticks axis fontsize: 14

Legend fontsize: 12

Width (px): 900

Height (px): 500

X-axis label: Time

Time unit: ☒ s ☐ min ☐ h ☐ Day ☐ Month ☐ Year

Right axis off ☒

Figure 7.1: Configuration area for selecting cases, model elements, represented parameters, display units and graphical settings.

Figure 7.2 illustrates the single-axis representation using the nominal temperature response of SOLAR_PANEL_1. The curves show the heating and cooling of the panel throughout the simulated orbital period, while their separation indicates the temperature range among its nodes.

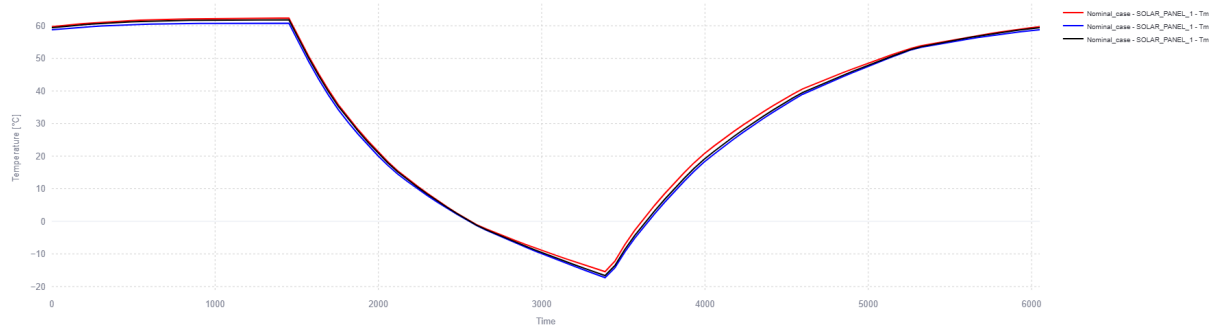


Figure 7.2: Minimum, mean and maximum temperatures of SOLAR_PANEL_1 for the nominal transient case.

A second vertical axis can be added when parameters with different units are combined in one plot. The parameters assigned to both axes retain a common time coordinate, allowing their evolution to be examined simultaneously.

Figure 7.3 illustrates a dual-axis representation for the battery. Its temperatures are displayed on the left axis, while the internal load Q_I is shown on the right axis. The values assigned during and outside eclipse are given in Table 5.4.

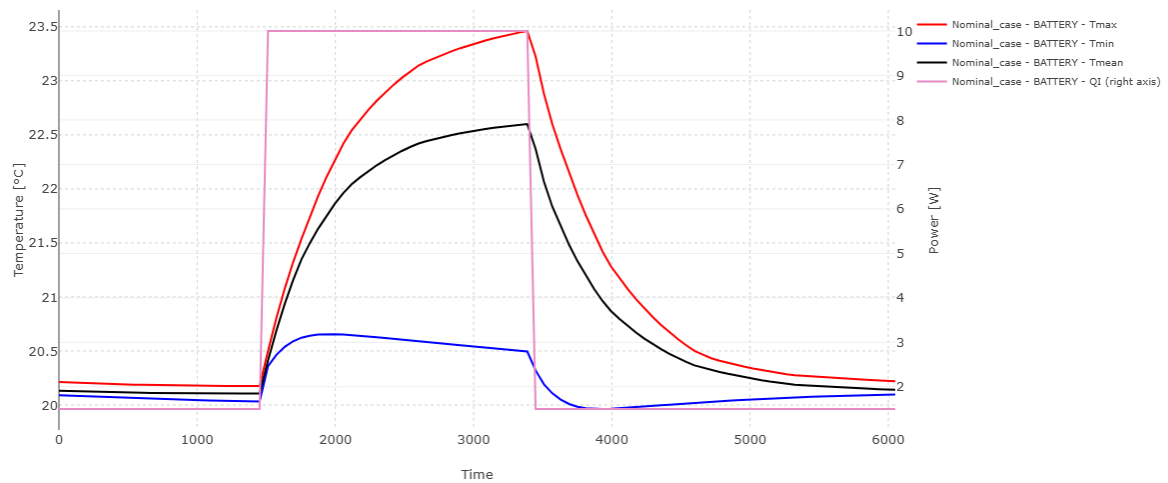


Figure 7.3: Battery temperatures and eclipse-dependent internal load Q_I for the nominal transient case.

7.2. Interface Temperature-Difference Analysis

The second area examines the temperature difference between two selected groups. The interface is defined by a source group and a target group, and several source–target pairs may be added to the same analysis. One or more cases can then be evaluated using the resulting interface selection.

The target list may be restricted to groups connected to the source through real GL or GR conductors. This filter facilitates the identification of interfaces represented directly in the thermal network. The user can add one target at a time or include all the connections detected for the chosen source.

Two calculation bases are available, together with an optional limit for the maximum permitted temperature difference. Figure 7.4 shows the controls used to select the cases, source and target groups, calculation method and acceptance limit.

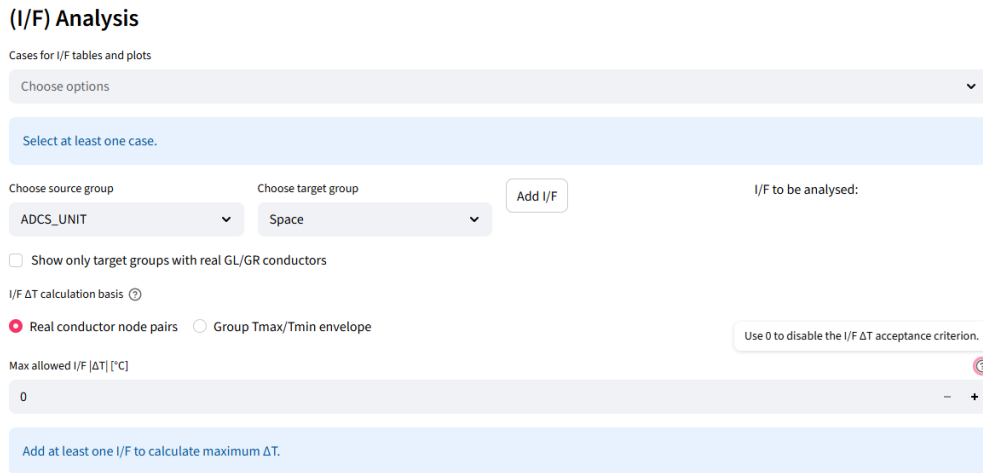


Figure 7.4: Configuration area for selecting cases, source and target groups, the interface calculation basis and the optional temperature-difference limit.

The default calculation basis uses the real node pairs connected through GL or GR conductors. Let $\mathcal{P}_{AB}$ contain the pairs (i, j) linking source group A with target group B . The absolute temperature difference associated with each pair is

$$\Delta T_{ij}(t) = |T_i(t) - T_j(t)| \quad (7.5)$$

At every output instant, the interface value corresponds to the largest difference among the connected pairs:

$$\Delta T_{A,B}^{I/F}(t) = \max[\Delta T_{ij}(t)]_{(i,j) \in \mathcal{P}_{AB}} \quad (7.6)$$

The maximum difference reached throughout the complete simulation is then

$$\Delta T_{A,B,\text{m}\acute{\text{a}}\text{x}}^{I/F} = \text{m}\acute{\text{a}}\text{x} \left[\Delta T_{A,B}^{I/F}(t) \right]_t \quad (7.7)$$

Once this result has been identified, the tool represents the complete temperature curves of the two nodes forming the critical pair. The hotter node and the colder node are displayed separately, while a vertical marker indicates the instant at which the maximum difference occurs.

The example presented below uses **OBC_BOX** as the source and **RADIATOR** as the target. This interface is representative because the thermal network includes an additional conductive link between both elements, as described in Section 5.4. Figure 7.5 shows that the maximum temperature difference reaches 21.4 °C at 3388.1 s.

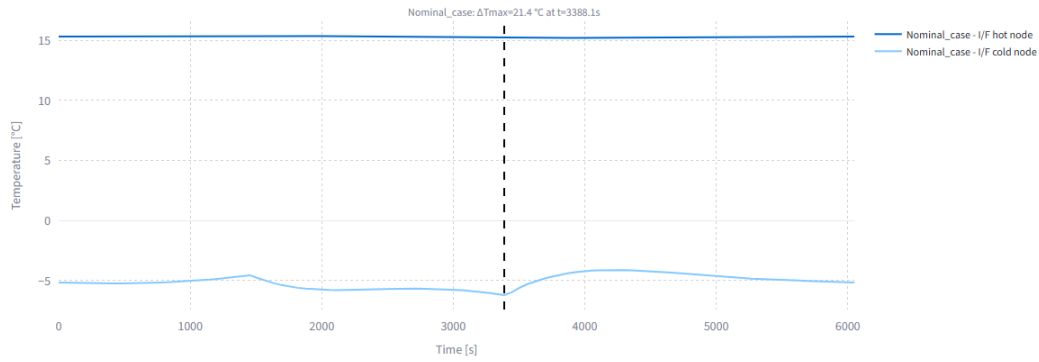


Figure 7.5: Hot-node and cold-node temperatures for the interface between **OBC_BOX** and **RADIATOR** in the nominal transient case. The vertical marker identifies the maximum temperature difference.

The same source group was subsequently used to add all the targets detected from its real conductors. For each source–target pair, Table 7.1 records the maximum absolute temperature difference, the event time, the hot and cold temperatures, the critical node pair and the coupling involved.

Table 7.1: Maximum interface temperature differences detected from **OBC_BOX** in the nominal transient case.

Target group	$ \Delta T _{\text{m}\acute{\text{a}}\text{x}}$ [°C]	Time [s]	T_{hot} [°C]	T_{cold} [°C]	Hot–cold nodes	Coupling
RADIATOR	21.41	3388.1	15.19	-6.23	31023–35506	GL+GR
BODY	7.80	5929.2	14.56	6.76	31014–11102	GL+GR
BODY_PLATE_MED	2.52	3388.1	15.49	12.97	12602–31011	GL+GR
BODY_PANEL	7.80	5929.2	14.56	6.76	31014–11102	GR
BODY_PLATE_SUP	2.96	1512.6	15.43	12.47	31022–13603	GR
Space	285.51	2299.1	15.51	-270.00	31022–99999	GR
BODY_PLATE_INF	6.04	1875.6	19.88	13.85	12000–31008	GR

Note: **Space** is included because the detection also identifies GR links to the environmental boundary. Identical results may occur when target groups overlap within the model hierarchy.

The alternative calculation basis uses the complete temperature envelopes of the source and target groups. It does not require the nodes producing the result to be connected by a real conductor. The corresponding difference is defined as

$$\Delta T_{A,B}^{\text{env}}(t) = \text{máx}[|T_{A,\text{máx}}(t) - T_{B,\text{mín}}(t)|, |T_{B,\text{máx}}(t) - T_{A,\text{mín}}(t)|] \quad (7.8)$$

This method provides a more conservative comparison between the complete groups, but the nodes defining the result do not necessarily represent a physical interface. The conductor-based calculation therefore offers greater traceability whenever a direct GL or GR connection is available.

A maximum permitted difference ΔT_{lim} can also be introduced. The available margin is calculated from

$$M_{\text{I/F}} = \Delta T_{\text{lim}} - \Delta T_{\text{máx}} \quad (7.9)$$

where $\Delta T_{\text{máx}}$ denotes the maximum value obtained with the selected calculation basis. A non-negative margin produces a **PASS** result, whereas a negative value produces **FAIL**. Setting ΔT_{lim} to zero disables this criterion, as in the nominal example presented above.

7.3. Temperature Variation over Selected Time Windows

The third area evaluates how much the temperature of each node changes over a selected time window. This representation complements the absolute temperatures examined in Section 7.1, since a component may remain within its operating limits while still undergoing a significant increase or decrease during a shorter interval.

The configuration starts with the selection of one or more cases, model elements and time windows. Several windows can be processed simultaneously, and the resulting variations may be represented through their maximum and minimum nodal values or supplemented with the corresponding mean curve. The elapsed-time unit, graph dimensions, axis labels and font sizes can also be modified. Two optional limits evaluate the maximum absolute temperature change and its average magnitude over the selected interval.

Figure 7.6 shows the controls used to define the cases, elements, time windows, representation method and acceptance criteria.

Temporal evolution of ΔT for different Δt windows

Represents $\Delta T(t) = T(t) - T(t - \Delta t)$ for the selected elements and for different time windows Δt .

Cases

Choose options

Elements

ADCS_UNIT x

Δt

Choose options

Select at least one case.

Ticks axis fontsize

14 - +

Legend fontsize

12 - +

Width (px)

900 - +

Height (px)

500 - +

X-axis label

Time

Time unit

☒ s
 ☐ min
 ☐ h

☐ Day
 ☐ Month

☐ Year

ΔT representation

Max/Min envelo...

Case representation

Overlay cases

☐ Show ΔT table

Max allowed $|\Delta T|$ [°C]

0 - +

Max allowed $|\Delta T| / \Delta t$ [°C/s]

0 - +

Select at least one case.

Figure 7.6: Configuration area for selecting the cases, model elements, time windows, representation method and optional temperature-variation limits.

For a requested window τ , the temperature variation of node n is calculated from

$$\Delta T_n(t; \tau) = T_n(t) - T_n(t - \tau^*) \quad (7.10)$$

where τ^* is the effective interval reproduced by the output-time vector. When the simulation uses a uniform output step Δt_m , the number of samples required for the calculation is

$$k_\tau = \left\lceil \frac{\tau}{\Delta t_m} \right\rceil \quad \tau^* = k_\tau \Delta t_m \quad (7.11)$$

The ceiling operation prevents the effective interval from being shorter than the requested value. If both values differ, the interface reports τ^* together with the selected window.

For a group g , the maximum–minimum representation is obtained from the nodal variations:

$$\Delta T_{g,\min}(t; \tau) = \min[\Delta T_n(t; \tau)]_{n \in \mathcal{N}_g} \quad (7.12)$$

$$\Delta T_{g,\max}(t; \tau) = \max[\Delta T_n(t; \tau)]_{n \in \mathcal{N}_g} \quad (7.13)$$

Positive values indicate that the node has warmed during the preceding interval, whereas negative values indicate cooling. The separation between both curves shows whether the nodes of the selected

element follow a similar evolution or experience different temperature changes.

The nominal example uses BODY_PANEL and requested windows of 10 and 30 minutes. This element contains several nodes with sufficiently different responses to make the maximum–minimum envelopes visible. The effective intervals obtained from the available output grid are approximately 10.084 and 30.251 minutes.

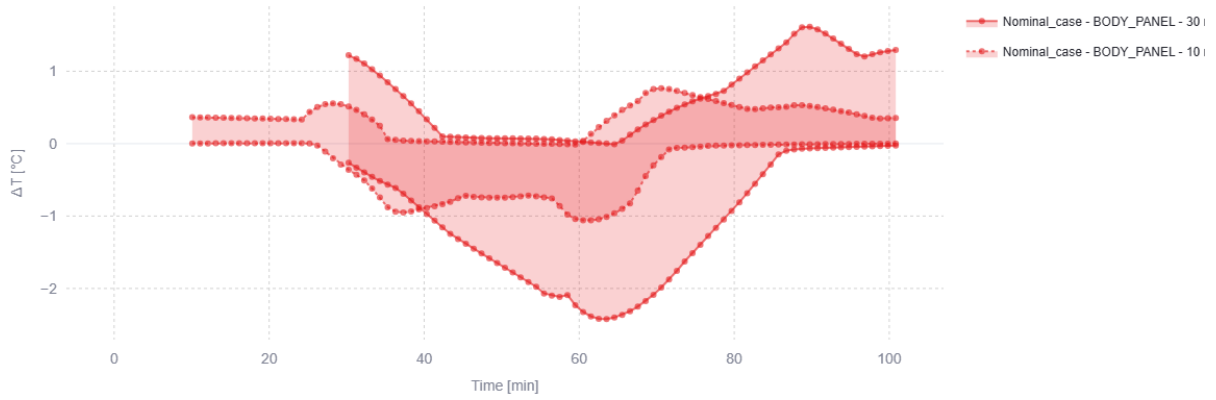


Figure 7.7: Maximum and minimum temperature variations of BODY_PANEL over 10- and 30-minute windows in the nominal transient case.

The longer window produces a wider envelope because it accumulates the temperature change over a greater part of the orbital cycle. However, the largest absolute variation does not by itself indicate how rapidly the change occurs. For this purpose, the application also calculates the average magnitude over the effective interval:

$$R_{T,n}(t; \tau) = \frac{|\Delta T_n(t; \tau)|}{\tau^*} \quad (7.14)$$

This parameter represents an average change over the selected window rather than an instantaneous derivative. Its value therefore depends on the duration of the interval.

Table 7.2 summarises the critical results obtained for both windows. It includes the minimum and maximum temperature changes, their event times and nodes, the largest absolute variation and its corresponding average magnitude.

Table 7.2: Temperature variations of BODY_PANEL over the selected time windows in the nominal transient case.

Requested window (min)	Effective window (min)	$\Delta T_{\min}$ (°C)	$t_{\min}$ (min)	Critical node	$\Delta T_{\max}$ (°C)	$t_{\max}$ (min)	Critical node	$ \Delta T _{\max}$ (°C)	$ \Delta T _{\max}/\tau^*$ (°C/min)
10	10.084	-1.06	61.51	10202	0.77	70.59	10602	1.06	0.105
30	30.251	-2.42	63.53	10202	1.61	89.75	10602	2.42	0.080

The 30-minute interval reaches the largest absolute variation, 2.42 °C, whereas the 10-minute interval produces the higher average magnitude, 0.105 °C/min. The two results therefore describe different aspects of the transient response: the longer window identifies the greater accumulated change, while the shorter one is more sensitive to rapid variations.

The complete nodal values can also be displayed and exported from the interface. Optional limits for $|\Delta T|$ and $|\Delta T|/\tau^*$ allow margins and acceptance states to be calculated. A value of zero disables the corresponding criterion, as in the example presented here.

7.4. Transient Thermal Summary Table

The fourth area condenses the transient temperature results of the selected elements into a single tabular output. Instead of representing their complete evolution, it identifies the main temperature extrema and the largest and smallest spatial temperature ranges reached during the simulation. This provides an initial overview of the most relevant values before examining a specific element in greater detail.

The configuration requires one or more cases and model elements. The case representation selector also enables the direct and difference-based options introduced in Section 6.4.2. The nominal example uses a single case, since this is sufficient to explain the content and organisation of the resulting table.

Figure 7.8 shows the selection area of this function.

Thermal summary table ⌵

Cases

Choose options ⌵

Select at least one case.

Elements

ADCS_UNIT ×

Select at least one element.

Case representation

Overlay cases ⌵

Select at least one case for the summary table.

Figure 7.8: Configuration area for selecting the cases and model elements included in the transient thermal summary.

Let $\mathcal{T}_{\text{out}}$ denote the set of output instants contained in the transient result file. The global maximum and minimum temperatures of a group g are calculated as

$$T_{g,\text{máx}}^{\text{glob}} = \max_{\substack{n \in \mathcal{N}_g \\ t \in \mathcal{T}_{\text{out}}}} [T_n(t)] \quad (7.15)$$

$$T_{g,\text{mín}}^{\text{glob}} = \min_{\substack{n \in \mathcal{N}_g \\ t \in \mathcal{T}_{\text{out}}}} [T_n(t)] \quad (7.16)$$

For each extreme, the output also identifies the node and simulation time at which it occurs. These data allow the critical value to be located without inspecting the temperature curves of all the nodes individually.

The table also evaluates the spatial temperature range of each group at every output instant:

$$\Delta T_g^{\text{sp}}(t) = T_{g,\text{máx}}(t) - T_{g,\text{mín}}(t) \quad (7.17)$$

Its maximum and minimum values throughout the simulation are then obtained from

$$\Delta T_{g,\text{máx}}^{\text{sp}} = \max_{t \in \mathcal{T}_{\text{out}}} [\Delta T_g^{\text{sp}}(t)] \quad (7.18)$$

$$\Delta T_{g,\text{mín}}^{\text{sp}} = \min_{t \in \mathcal{T}_{\text{out}}} [\Delta T_g^{\text{sp}}(t)] \quad (7.19)$$

The corresponding times and hot–cold node pairs are retained in the output. This calculation differs from the time-window method of Section 7.3. There, $\Delta T_n(t; \tau)$ compares one node at two different instants, whereas $\Delta T_g^{\text{sp}}(t)$ compares different nodes of the same element at one instant.

The original output contains the temperature extrema and spatial ranges in a single table. For readability in the document, the nominal results are divided into two complementary tables. The first contains the global maximum and minimum temperatures, together with their event times and critical nodes, while the second contains the maximum and minimum spatial ranges reached by the same elements. The associated node pairs identify the locations that define each range.

Table 7.3: Global temperature extrema of the selected elements in the nominal transient case.

Element	$T_{\max}$ [°C]	$t_{T_{\max}}$ [s]	Node	$T_{\min}$ [°C]	$t_{T_{\min}}$ [s]	Node
ADCS_UNIT	18.81	2964.6	33002	18.04	4840.2	33005
SOLAR_PANEL_2	62.29	1452.1	73004	-17.34	3388.1	75203
SOLAR_PANEL_1	62.35	1452.1	70004	-17.27	3388.1	72502
MLI_PANELS	134.92	1452.1	50800	-107.97	4356.2	51003
COMMS_BOX	20.81	2057.1	32000	19.24	3630.1	32005
OBC_BOX	15.51	2299.1	31022	12.97	3388.1	31011
RADIATOR	1.33	4235.2	35504	-6.23	3388.1	34002
BATTERY	23.46	3388.1	30000	19.96	3932.7	30005
CAMERA_CYL	-1.76	1452.1	42500	-27.80	3388.1	43500
CAMERA_ELEC	12.78	2783.1	40501	12.17	0.0	40500
CAMERA_BASE	-6.57	1452.1	40404	-15.39	3630.1	40000
BODY_PLATE_INF	20.25	1694.1	12001	17.85	4537.7	12502
BODY_PLATE_MED	15.57	2359.6	13002	14.98	3690.7	12601
BODY_PLATE_SUP	12.76	2178.1	13600	12.34	4961.2	13103
BODY_PANEL	20.56	1452.1	10601	6.66	4053.7	11002

Table 7.4: Maximum and minimum spatial temperature ranges of the selected elements in the nominal transient case.

Element	$\Delta T_{\max}^{\text{sp}}$ (°C)	$t_{\Delta T_{\max}}$ [s]	Hot-cold nodes	$\Delta T_{\min}^{\text{sp}}$ (°C)	$t_{\Delta T_{\min}}$ [s]	Hot-cold nodes
ADCS_UNIT	0.62	4053.7	33002-33005	0.49	1754.6	33002-33005
SOLAR_PANEL_2	2.62	3811.7	73005-75200	0.15	2541.1	73004-75203
SOLAR_PANEL_1	2.65	3811.7	70003-72502	0.20	2541.1	70003-72500
MLI_PANELS	240.79	4053.7	50801-51003	111.98	2722.6	50500-51003
COMMS_BOX	1.19	3448.6	32000-32005	0.45	1452.1	32002-32005
OBC_BOX	2.47	3388.1	31022-31011	1.17	5505.7	31022-31011
RADIATOR	5.82	786.5	35504-34002	5.48	4295.7	35504-34002
BATTERY	2.97	3388.1	30000-30005	0.12	484.0	30002-30004
CAMERA_CYL	0.00	3448.6	42500-43500	0.00	1815.1	42500-43500
CAMERA_ELEC	0.45	1633.6	40501-40500	0.44	3932.7	40501-40500
CAMERA_BASE	7.41	1452.1	40404-40008	5.06	1936.1	40009-40008
BODY_PLATE_INF	2.19	1633.6	12001-12502	1.70	3630.1	12001-12502
BODY_PLATE_MED	0.48	3085.6	13002-12601	0.30	4416.7	13000-12601
BODY_PLATE_SUP	0.26	5203.2	13600-13103	0.16	3267.1	13600-13103
BODY_PANEL	13.71	1452.1	10601-11002	12.73	3388.1	10701-11002

The two outputs serve different purposes. Global extrema indicate the highest and lowest temperatures reached by an element during the complete simulation, while the spatial ranges describe its internal thermal non-uniformity at a given instant. Their combined use therefore distinguishes a general shift in component temperature from a large difference between its individual nodes.

The summary can be generated for any selection of elements and exported for later inspection. No separate graph is required in this area, since its main function is to provide a compact numerical screening of the transient solution.

7.5. Conductive and Radiative Heat-Flow Evolution

The fifth function examines the heat transferred between a selected source and target throughout the transient simulation. After choosing one or more cases, the user defines both elements and selects the conductive, radiative or total heat-flow parameters to be represented. The output can be expressed in watts or milliwatts, while the horizontal axis uses the selected elapsed-time unit.

Figure 7.9 shows the corresponding controls. The three available parameters are Q_{GL} , Q_{GR} and Q_{GT} , representing the linear conductive, radiative and combined exchanges, respectively. Their sign follows the source–target order defined in the interface.

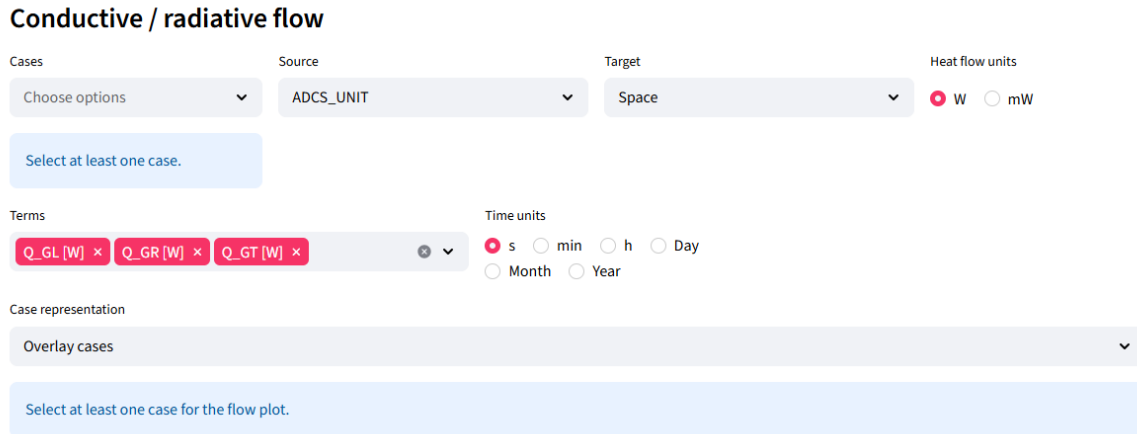


Figure 7.9: Configuration area for selecting the cases, source and target elements, heat-flow parameters and display units.

Let $\mathcal{P}_{AB}^{GL}$ contain the node pairs connected by linear conductors between source A and target B . The conductive contribution is calculated as

$$Q_{GL,A \leftarrow B}(t) = \sum_{(i,j) \in \mathcal{P}_{AB}^{GL}} GL_{ij} [T_j(t) - T_i(t)] \quad (7.20)$$

where node i belongs to the source and node j to the target. The radiative contribution is obtained from the pairs contained in $\mathcal{P}_{AB}^{GR}$:

$$Q_{GR,A \leftarrow B}(t) = \sigma \sum_{(i,j) \in \mathcal{P}_{AB}^{GR}} GR_{ij} [T_{j,K}^4(t) - T_{i,K}^4(t)] \quad (7.21)$$

The temperatures are converted to kelvin before evaluating the fourth-power term. The total exchange combines both mechanisms:

$$Q_{GT,A \leftarrow B}(t) = Q_{GL,A \leftarrow B}(t) + Q_{GR,A \leftarrow B}(t) \quad (7.22)$$

With this ordering, a positive value represents heat entering the selected source from the target. A negative value indicates heat leaving the source towards the target. Reversing the source and target therefore reverses the sign of the represented exchange.

The nominal example uses `BATTERY` as the source and `BODY_PLATE_INF` as the target. The three heat-flow parameters are represented in watts over the complete simulation. Figure 7.10 shows that all values remain negative, indicating a continuous transfer from the battery towards its supporting plate.

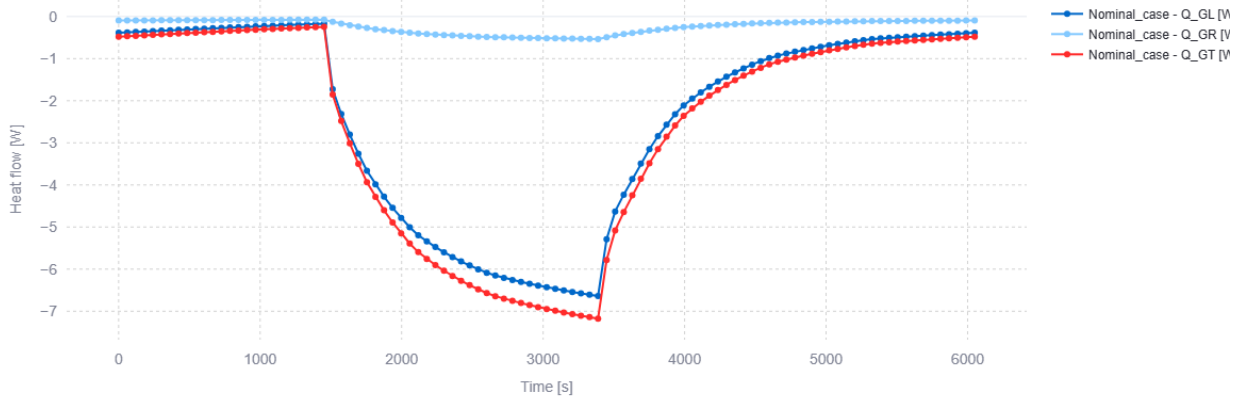


Figure 7.10: Conductive, radiative and total heat flows between `BATTERY` and `BODY_PLATE_INF` in the nominal transient case. Negative values indicate heat leaving the battery.

The conductive term provides the main contribution, while the radiative exchange remains comparatively small. Consequently, Q_{GT} follows the shape of Q_{GL} , with a slightly larger outward magnitude. The transfer increases during the eclipse interval and reaches its most negative value at approximately 3388.1 s.

The interface also generates a table containing the value of each parameter at every output instant. Since the complete export would be unnecessarily long for the document, Table 7.5 summarises only the minimum and maximum values and their corresponding times.

Table 7.5: Extreme conductive, radiative and total heat-flow values for the BATTERY-BODY_PLATE_INF selection in the nominal transient case.

Term	Minimum [W]	$t_{\min}$ [s]	Maximum [W]	$t_{\max}$ [s]
Q_{GL}	-6.641	3388.1	-0.170	1452.1
Q_{GR}	-0.535	3388.1	-0.074	1452.1
Q_{GT}	-7.176	3388.1	-0.243	1452.1

The total heat flow ranges from approximately -0.243 W to -7.176 W. The negative sign is retained in the table because it forms part of the source–target convention, rather than indicating the magnitude alone.

7.6. Absorbed and Released Thermal Energy

The final area converts the thermal-power parameters associated with a selected element into cumulative energy. Its configuration begins with the selection of one or more cases and a model element. The user then defines the power contributions to be included and chooses the time, power and energy units used in the representation.

The available parameters comprise Q_A , Q_E , Q_S , Q_I and Q_R , together with the net conductive and radiative exchanges with nodes outside the selected group, $Q_{GL,ext}$ and $Q_{GR,ext}$. The complete thermal balance is also available and is calculated as

$$Q_{balance,g}(t) = Q_{A,g}(t) + Q_{E,g}(t) + Q_{S,g}(t) + Q_{I,g}(t) + Q_{R,g}(t) + Q_{GL,ext,g}(t) + Q_{GR,ext,g}(t) \quad (7.23)$$

where the external conductive and radiative terms contain only exchanges between nodes belonging to group g and nodes outside that group. Internal connections between two nodes of the selected element are excluded because they redistribute energy within the group without changing its total balance.

Figure 7.11 shows the controls used to select the cases, element, power parameters and display units. The power curves may be expressed in watts or milliwatts, while the accumulated energy is represented in joules or watt-hours.

Absorbed/released energy of components ⇌

Cases
Choose options ▼
Select at least one case.

Element
ADCS_UNIT ▼

Time unit
☒ s ☐ min
☐ h ☐ Day
☐ Month ☐ Year

Energy unit
☒ J ☐ Wh

Power unit
☒ W ☐ mW

Case representation
Overlay cases ▼
Select at least one case.

Figure 7.11: Configuration area for selecting the cases, model element, thermal-power parameters and units used in the energy calculation.

For a selected parameter $Q_{x,g}$, the energy transferred from the start of the simulation to time t is

$$E_{x,g}(t) = \int_{t_0}^t Q_{x,g}(\xi) d\xi \quad (7.24)$$

The numerical integration follows the output-time vector contained in the result file. Between two successive instants, the trapezoidal rule gives

$$E_{x,g}(t_k) = E_{x,g}(t_{k-1}) + \frac{Q_{x,g}(t_{k-1}) + Q_{x,g}(t_k)}{2} (t_k - t_{k-1}) \quad (7.25)$$

with $E_{x,g}(t_0) = 0$. When time is expressed in seconds, the resulting energy is obtained directly in joules. Its conversion to watt-hours follows

$$E_{x,g}^{\text{Wh}}(t) = \frac{E_{x,g}^{\text{J}}(t)}{3600} \quad (7.26)$$

The sign of the power parameter determines the slope of its accumulated energy curve. A positive value increases the stored integral and represents energy entering the selected group, whereas a negative value decreases it and represents energy leaving the group. When the power becomes zero, the corresponding cumulative energy remains constant.

The nominal example uses the group **SOLAR_PANELS** and represents the albedo and direct-solar contributions Q_A and Q_S . Power is expressed in watts, energy in watt-hours and elapsed time in minutes. Both the instantaneous parameters and their integrated values are displayed in the same graph using separate vertical axes.

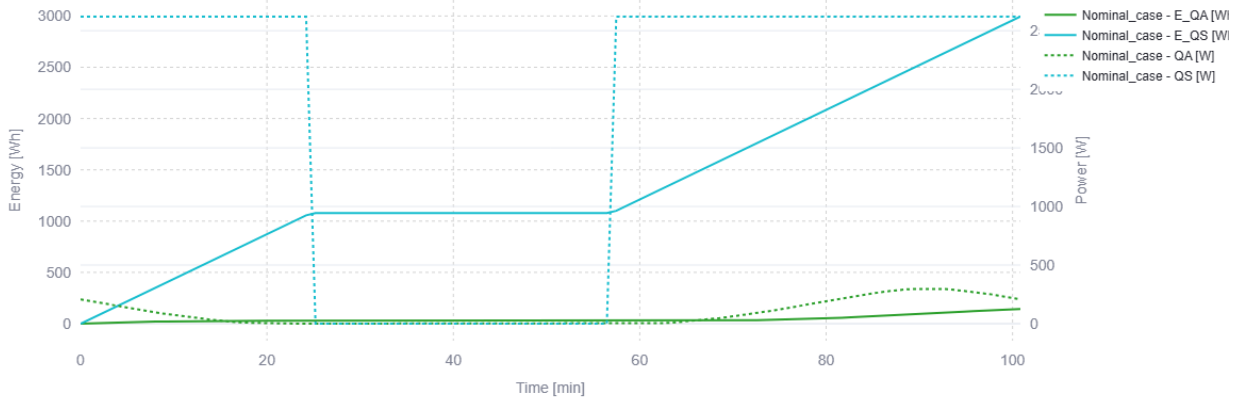


Figure 7.12: Albedo and direct-solar thermal powers and their cumulative energies for SOLAR_PANELS in the nominal transient case.

The direct-solar contribution dominates while the spacecraft remains illuminated and becomes zero during eclipse. Consequently, its cumulative energy increases during the illuminated intervals and remains constant while direct solar radiation is absent. The albedo contribution is smaller and varies according to the orbital position, producing a more gradual change in its integrated value.

The graph illustrates the relation between an instantaneous thermal input and the energy transferred over a finite interval. The same calculation can be applied to individual loads, external heat exchanges or the complete Q_{balance} parameter. In the latter case, increases and decreases of the cumulative curve identify periods of net energy absorption and release by the selected element.

These results describe thermal power and thermal energy obtained from the ESATAN-TMS solution. They do not represent the electrical energy generated by the solar cells. The conversion of the incident solar load into electrical power and its use within the battery model are developed in the following chapter.

8. Electrical Power Balance and Battery Model

The **Energy balance** tab extends the thermal post-processing workflow with a separate electrical calculation. The module first uses the solar load exported by ESATAN-TMS to estimate the power generated by the solar cells. It then obtains the electrical demand from the internal dissipations of the selected equipment and applies the resulting power surplus or deficit to a configurable lithium-ion battery model.

All operations take place after the thermal solution. The application reads Q_S , Q_I and the required temperature histories from the imported .TMD file, but it does not modify those data or return any electrical quantity to ESATAN-TMS. The thermal and electrical calculations therefore remain one-way coupled: the solved thermal case supplies the inputs, and the Python module derives solar generation, electrical demand, battery state and operating limits.

The workflow contains two consecutive stages. The first converts the selected solar-panel result into $Q_{S,\text{elec}}$ and builds Q_{load} . The second uses both histories, together with the battery temperature and the configurable cell parameters, to calculate stored energy, state of charge, terminal voltage, charge and discharge currents, losses and operating phases.

8.1. Solar-to-Electrical Power Conversion

8.1.1. Conversion Method

The selected solar-panel group provides the absorbed solar load $Q_S(t)$. In the adopted thermo-electrical formulation, the incident solar power $Q_{S,i}(t)$ divides into a thermal fraction and an electrical fraction. The solar absorptivity α describes the total fraction retained by the solar cells, while the electrical conversion efficiency $\eta(t)$ defines the part converted into electrical power. The effective thermal absorptivity becomes

$$\alpha_{\text{th}}(t) = \alpha - \eta(t). \quad (8.1)$$

The thermal load exported for the selected group follows

$$Q_S(t) = [\alpha - \eta(t)] Q_{S,i}(t). \quad (8.2)$$

The application therefore reconstructs the incident power from

$$Q_{S,i}(t) = \frac{Q_S(t)}{\alpha - \eta(t)}, \quad (8.3)$$

and calculates the electrical contribution as

$$Q_{S,\text{elec}}(t) = \eta(t)Q_{S,i}(t) = \frac{\eta(t)}{\alpha - \eta(t)}Q_S(t). \quad (8.4)$$

The calculation requires $0 < \eta(t) < \alpha \leq 1$ throughout the simulated interval. The user may introduce η as a constant or as a function of the mean temperature of the selected solar-panel element. A temperature-dependent expression therefore has the general form

$$\eta(t) = f_\eta[T_{\text{SP,mean}}(t)]. \quad (8.5)$$

This option allows the conversion efficiency to follow the thermal result without requiring a complete current–voltage or maximum-power-point model. When the user enters a constant, the same value applies to every output instant.

The electrical demand comes from the internal dissipations of the selected consumer groups. For the set of load elements $\mathcal{G}_{\text{load}}$, the application calculates

$$Q_{\text{load}}(t) = \sum_{g \in \mathcal{G}_{\text{load}}} \max[Q_{I,g}(t), 0]. \quad (8.6)$$

This approximation treats the positive internal load of each selected unit as its electrical consumption. It suits the electronic equipment represented in the model, for which most of the consumed power finally appears as heat. The method does not separately account for exported mechanical work, radio-frequency power or conversion losses outside the battery model.

The difference between generation and demand defines the power available before battery operation:

$$P_{\text{net}}(t) = Q_{S,\text{elec}}(t) - Q_{\text{load}}(t). \quad (8.7)$$

Positive values provide solar surplus for charging, whereas negative values require battery discharge. The corresponding non-negative magnitudes are

$$P_{\text{sur}}(t) = \max[P_{\text{net}}(t), 0], \quad (8.8)$$

$$P_{\text{def}}(t) = \max[-P_{\text{net}}(t), 0]. \quad (8.9)$$

Solar generation always supplies the load first. The battery only receives the remaining surplus or

covers the remaining deficit.

8.1.2. Input Selection and Electrical-Balance Display

The upper area of the tab gathers the inputs required by the previous relations. The user first selects a transient case and defines the efficiency through the η *equation or constant* field. When the expression contains T , the application evaluates it with the mean temperature of the selected solar-panel element in degrees Celsius. The solar conversion does not use terminal voltage as an efficiency variable.

The *Elements* selector lists the hierarchy groups with a non-zero Q_S history. The chosen group identifies the solar-panel nodes whose load enters Equation (8.3). The Q_{load} *elements* selector then lists the groups with a non-zero Q_I contribution. The application sums every selected group, so the user must avoid choosing a parent group together with one of its descendants when both contain the same nodes. Such a selection would count the same internal dissipation twice.

The thermo-optical selector retrieves the solar absorptivity from the property sets contained in the imported results. When the required set is unavailable or the user needs another value, the interface accepts manual values of α and ε . The current conversion uses α ; ε remains stored with the analysis to preserve the selected thermo-optical definition.

Figure 8.1 shows the configuration area. The layout follows the calculation sequence: efficiency, solar element, electrical-load elements, thermo-optical properties and output curves.

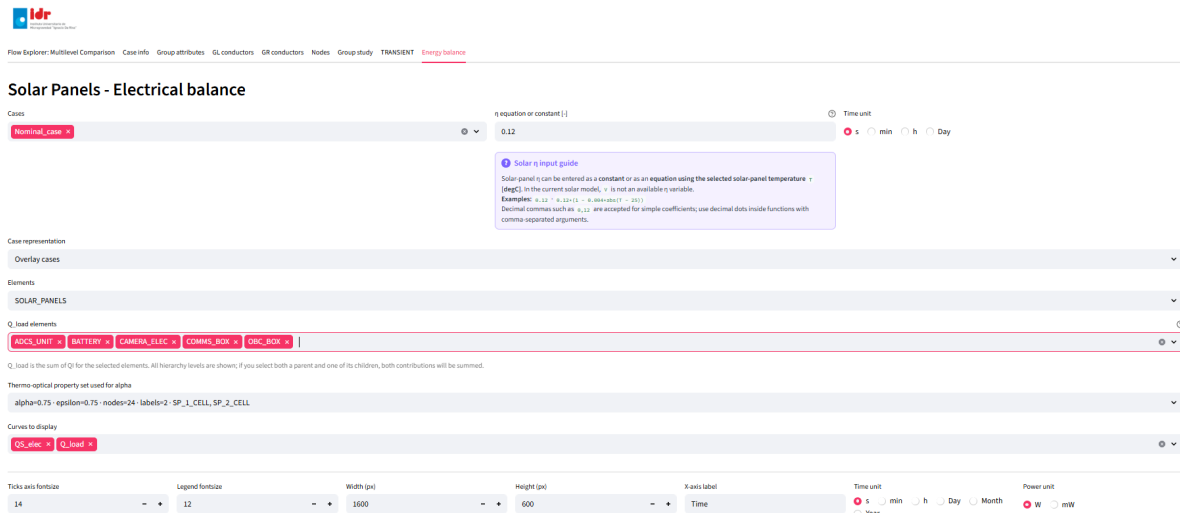


Figure 8.1: Configuration of the solar-to-electrical conversion and the electrical-load selection.

The graph can display Q_S , $Q_{S,elec}$ and Q_{load} on the same time axis. The default selection compares the available electrical generation with the estimated demand, which directly reveals the charging and discharging intervals later used by the battery model. Figure 8.2 illustrates this output. Its

numerical values only demonstrate the representation available in the tab; the final case results belong to the subsequent results analysis.

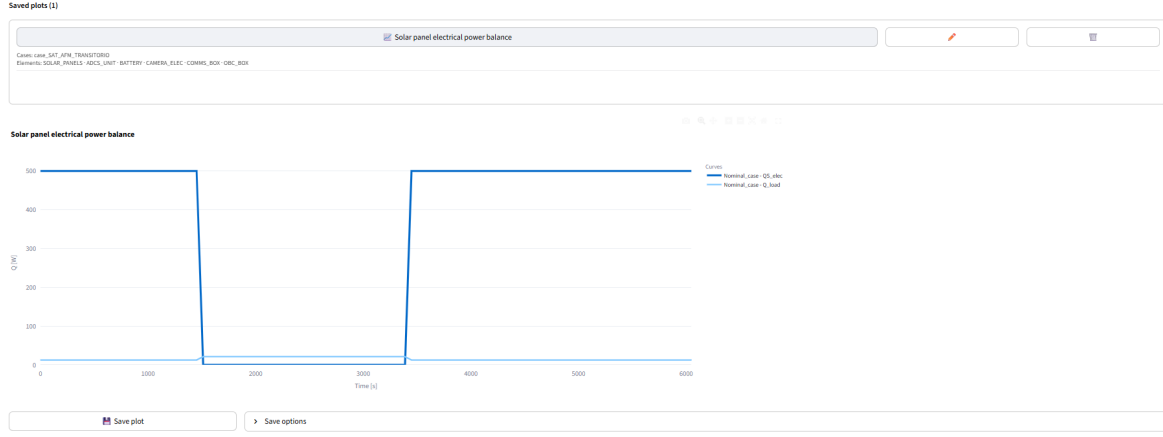


Figure 8.2: Illustrative solar-panel electrical balance containing $Q_{S,elec}$ and Q_{load} .

An optional calculation table retains the values used at every output instant. Its columns include Q_S , $Q_{S,i}$, $Q_{S,elec}$, Q_{load} , α , α_{th} and η . When the efficiency depends on temperature, the table also includes the temperature history supplied to the efficiency expression. This output allows the user to check the complete conversion before applying the result to the battery.

8.2. Battery Model

The battery calculation starts from the histories generated in the first part of the tab. At each time interval, solar power first covers the selected load. The model sends any remaining surplus to the battery and requests discharge when generation cannot meet the demand. Battery temperature, conversion efficiencies, voltage limits, internal resistance and charge-control settings then determine the accepted power and the evolution of the stored energy.

The application recalculates the battery whenever the user changes the solar element, the Q_{load} groups, the thermo-optical property, α or the solar-efficiency equation. Consequently, two identical battery configurations can produce different responses when their upstream $Q_{S,elec}$ or Q_{load} histories differ.

8.2.1. Energy-Discharge-Level Basis

The battery formulation takes the energy-discharge-level approach proposed by Porrás-Hermoso, Pindado and Cubas as its theoretical basis [19]. Their equivalent-circuit model uses discharged energy ϕ as the main state variable instead of relying only on discharged ampere-hours. This choice relates the state directly to the exchanged power and reduces the dependence of the capacity

description on a single current rate.

For a discharge process, the original formulation defines

$$\phi(t) = \phi_0 + \int_{t_0}^t [VI_d + R_d I_d^2] dt, \quad (8.10)$$

whereas charging decreases the discharged-energy level according to

$$\phi(t) = \phi_0 - \int_{t_0}^t [VI_c - R_c I_c^2] dt. \quad (8.11)$$

Here, I_d and I_c denote the non-negative discharge and charge current magnitudes, while R_d and R_c represent the corresponding internal resistances. The terminal voltage follows

$$V_d(\phi, I_d) = E_d(\phi) - R_d I_d, \quad (8.12)$$

$$V_c(\phi, I_c) = E_c(\phi) + R_c I_c. \quad (8.13)$$

Porras-Hermoso et al. identify $E_d(\phi)$ and $E_c(\phi)$ from experimental charge and discharge curves and extend the static equivalent circuit with additional resistor–capacitor branches for dynamic operation [19].

The implementation developed for this project retains the energy state, the separate charge and discharge resistances, and the terminal-voltage relations. It does not reproduce the experimentally fitted voltage functions or the dynamic resistor–capacitor branches. Instead, it constructs a generic piecewise-linear internal-voltage curve $E(\phi)$ from the cell discharge cut-off, nominal and charge-limit voltages. This reduction keeps the model configurable when no calibration data exist for a particular flight battery.

Let $C_{\text{pack,Wh}}$ denote the nominal energy capacity and $E_{\text{st}}(t)$ the stored energy. The implemented discharged-energy state becomes

$$\phi(t) = C_{\text{pack,Wh}} - E_{\text{st}}(t), \quad (8.14)$$

and the reported state of charge follows

$$\text{SOC}(t) = 100 \frac{E_{\text{st}}(t)}{C_{\text{pack,Wh}}}. \quad (8.15)$$

The module solves the corresponding relation $P = VI$ to obtain current from the power requested at each step. It also adds conversion efficiencies, charge and discharge power limits, temperature-dependent charge derating and a simplified CC–CV strategy. The resulting curves provide enginee-

ring estimates for the selected configuration rather than an experimentally calibrated prediction of a specific battery.

8.2.2. Thermal Derating and Battery-Pack Configuration

The first parameter block links the battery calculation with the thermal solution and defines the physical pack. The *Battery TMD element* selector identifies the group whose mean temperature represents the battery. The model uses this history in the charge-derating calculation and in any charge or discharge efficiency expression that contains T .

The lower and upper derating limits define the temperature interval in which the battery accepts charge. A common ramp margin m_T softens both transitions. For non-overlapping lower and upper ramps, the implemented factor can be written as

$$f_T(T) = \begin{cases} 0, & T \leq T_{\text{low}}, \\ \frac{T - T_{\text{low}}}{m_T}, & T_{\text{low}} < T < T_{\text{low}} + m_T, \\ 1, & T_{\text{low}} + m_T \leq T \leq T_{\text{high}} - m_T, \\ \frac{T_{\text{high}} - T}{m_T}, & T_{\text{high}} - m_T < T < T_{\text{high}}, \\ 0, & T \geq T_{\text{high}}. \end{cases} \quad (8.16)$$

The factor multiplies the maximum charge current and charge power. A value of one allows normal charging, a value of zero blocks it, and intermediate values reduce the accepted surplus linearly. The derating does not limit discharge in the current implementation.

The cell configuration uses N_s cells in series and N_p parallel strings. The application obtains the equivalent pack voltages from

$$V_{x,\text{pack}} = N_s V_{x,\text{cell}}, \quad x \in \{\text{cut}, \text{nom}, \text{chg}\}, \quad (8.17)$$

while the charge and nominal energy capacities become

$$C_{\text{pack,Ah}} = N_p C_{\text{cell,Ah}}, \quad (8.18)$$

$$C_{\text{pack,Wh}} = V_{\text{nom,pack}} C_{\text{pack,Ah}}. \quad (8.19)$$

The initial SOC fixes the energy available at the first output instant:

$$E_{st,0} = \frac{SOC_0}{100} C_{pack,Wh}. \quad (8.20)$$

Table 8.1 summarises every field in this first parameter block.

Table 8.1: Battery thermal and pack-configuration inputs.

Interface field	Role in the calculation
Battery TMD element	Supplies the mean battery temperature used by charge derating and temperature-dependent efficiency expressions.
Charge derating lower limit	Blocks charging below the selected lower temperature.
Charge derating upper limit	Blocks charging above the selected upper temperature.
Charge derating ramp margin	Defines the linear transition band beside both charging limits.
Series cells N_s	Scales the pack cut-off, nominal and charge-limit voltages.
Parallel strings N_p	Scales pack charge capacity and reduces equivalent resistance.
Cell charge capacity	Defines the ampere-hour capacity of one cell and, together with N_p , the pack charge capacity.
Cell discharge cut-off voltage	Defines the minimum admissible cell voltage during discharge.
Cell nominal/reference voltage	Defines the nominal pack voltage and the conversion from Ah to Wh.
Cell charge voltage limit	Defines the upper terminal-voltage limit used during charging.
Initial SOC	Sets the stored energy at the start of the transient calculation.

Figure 8.3 shows the corresponding controls in the application.

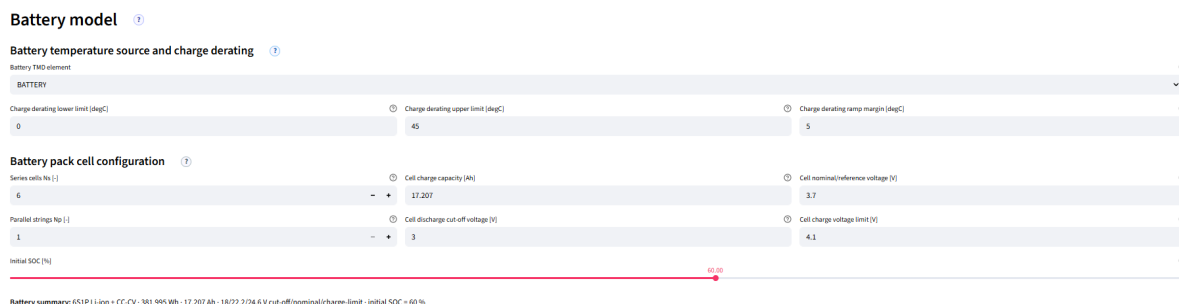


Figure 8.3: Battery-temperature reference, charge derating and cell-pack configuration.

8.2.3. Electrical Parameters and CC–CV Charge Control

The second parameter block defines the power-transfer behaviour of the pack. Charge and discharge efficiencies may use constants or expressions of battery temperature T and terminal voltage V :

$$\eta_c(t) = f_c[T_{batt}(t), V(t)], \quad \eta_d(t) = f_d[T_{batt}(t), V(t)]. \quad (8.21)$$

The application requires $0 < \eta_c \leq 1$ and $0 < \eta_d \leq 1$. Constants form the simplest valid expressions. During charge, η_c reduces the solar surplus that reaches the battery terminals. During discharge, η_d reduces the terminal power delivered to the electrical load. The model reports the remaining difference as conversion loss.

Separate cell resistances retain the charge–discharge asymmetry of the energy-based equivalent circuit. Their pack values follow

$$R_{x,\text{pack}} = R_{x,\text{cell}} \frac{N_s}{N_p}, \quad x \in \{d, c\}. \quad (8.22)$$

The resistance lowers the discharge terminal voltage and raises the charge terminal voltage. It also produces the internal loss

$$P_{\text{loss,int}}(t) = RI^2(t). \quad (8.23)$$

Maximum charge and discharge powers establish independent caps for both operating directions. During charging, the CC current ceiling follows the selected C-rate:

$$I_{\text{CC,max}} = C_{\text{rate}} C_{\text{pack,Ah}}. \quad (8.24)$$

The temperature factor from Equation (8.16) reduces this ceiling and the maximum charge power. The voltage constraint also requires

$$E(\phi) + R_c I_c \leq V_{\text{chg,pack}}. \quad (8.25)$$

Above the selected SOC threshold SOC_{CV} , an additional factor progressively reduces the current:

$$f_{\text{CV}}(\text{SOC}) = \left[\frac{100 - \text{SOC}}{100 - \text{SOC}_{\text{CV}}} \right]^\gamma, \quad \text{SOC} \geq \text{SOC}_{\text{CV}}, \quad (8.26)$$

where γ controls the shape of the reduction. The current remains subject to the CC limit below this threshold, provided that the solar surplus and charge-power cap allow it. The model declares practical full charge when the tapered or voltage-limited current falls below the selected current threshold; any additional surplus then appears as curtailed solar power.

During discharge, the model limits current through the available stored energy, the maximum discharge power and the cell discharge cut-off voltage. Any demand that remains after these restrictions appears as unserved load.

Table 8.2 collects every field in this second parameter block.

Table 8.2: Battery electrical and CC–CV control inputs.

Interface field	Role in the calculation
Charge efficiency equation	Defines the fraction of solar surplus transferred towards battery charging; it may depend on T and V .
Discharge efficiency equation	Defines the fraction of terminal discharge power delivered to the selected electrical load; it may depend on T and V .
Maximum charge power	Caps the terminal power accepted from the solar surplus.
Maximum discharge power	Caps the terminal power extracted when generation does not cover the demand.
Maximum constant-current charge rate	Sets the CC current ceiling as a multiple of pack capacity in Ah.
SOC where current reduction towards CV begins	Defines the SOC threshold above which the taper factor acts.
Current-reduction shape factor	Controls how sharply charge current decreases near full charge.
Charge-complete current threshold	Stops charging when the reduced current falls below the selected practical limit in the CV region.
Cell discharge internal resistance	Determines pack voltage drop and I^2R loss during discharge.
Cell charge internal resistance	Determines pack voltage rise and I^2R loss during charge.

Figure 8.4 shows these controls. The summary box below the inputs reports the resulting voltage limits, equivalent pack resistances and an efficiency preview for the current configuration.

Battery electrical parameters

Charge efficiency equation []

0.98

Discharge efficiency equation []

0.98

Max charge power [W]

357.24328

Max discharge power [W]

84.000001

CC-CV charge control and internal resistance

Maximum constant-current charge rate [C-rate]

1

SOC where current reduction towards CV begins [%]

85

Current-reduction shape factor near full charge []

1.5

Charge-complete current threshold [A]

0.1

Cell discharge internal resistance $R_{dis, cell}$ [ohm]

0.016666667

Cell charge internal resistance $R_{chg, cell}$ [ohm]

0.016666667

Voltage limits: cut-off = 10 V · nominal / ref = 22.2 V · charge limit = 24.6 V

Pack resistance: $R_{dis} = 0.1 \Omega$ · $R_{chg} = 0.1 \Omega$

Efficiency preview: $T = 20^\circ\text{C}$ · $V = 22.2 \text{ V}$ · $\eta_{a, charge} = 0.98$ · $\eta_{a, discharge} = 0.98$

Figure 8.4: Battery efficiencies, power limits, internal resistances and simplified CC–CV control parameters.

8.2.4. Battery Output Curves and Summary Results

The tab places the graphical outputs immediately after the parameter blocks so that the user can inspect the effect of each change without leaving the battery workflow. The first graph shows state of charge throughout the simulation. Rising segments indicate net storage, falling segments indicate battery support of the load, and horizontal segments identify intervals in which the stored energy remains unchanged or reaches an operating limit.

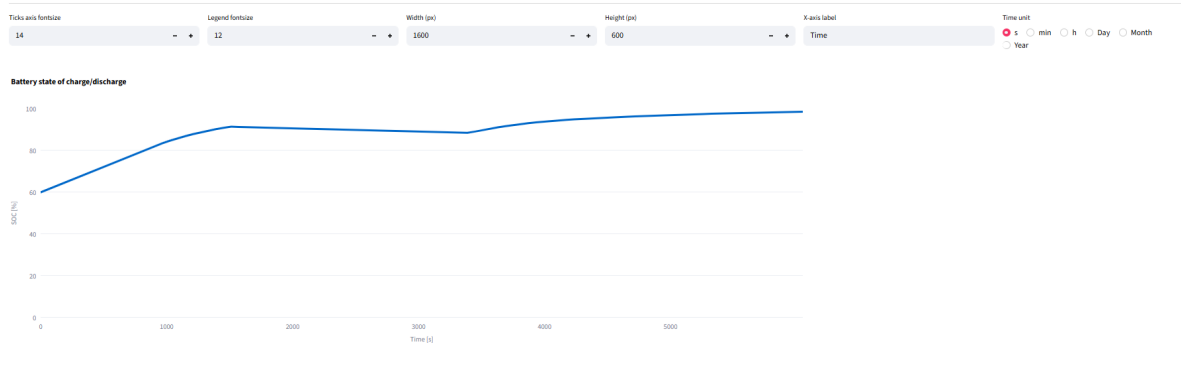


Figure 8.5: Illustrative battery state-of-charge history.

The second graph combines terminal voltage with separate non-negative charge and discharge currents. The left axis represents voltage, while the right axis represents current. The separation of both current directions avoids relying on a sign convention in the graphical interpretation, although the detailed calculation table also retains a signed battery current that is positive in discharge and negative in charge.

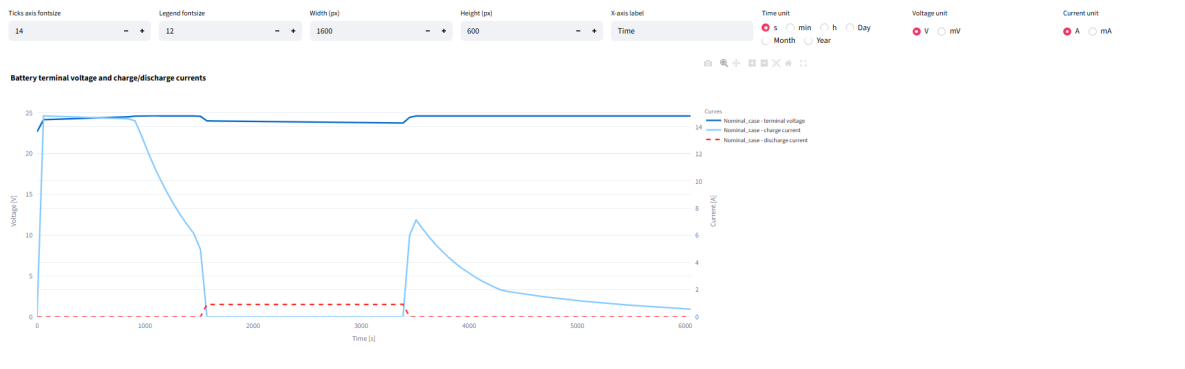


Figure 8.6: Illustrative terminal-voltage and charge/discharge-current histories.

The third graph connects the thermal result with the battery restrictions. It places the mean battery temperature on the left axis and the charge-derating, charge-efficiency and discharge-efficiency factors on the right. This output shows whether a change in battery behaviour comes from the available power, the selected efficiency expressions or a temperature-dependent charging restriction.

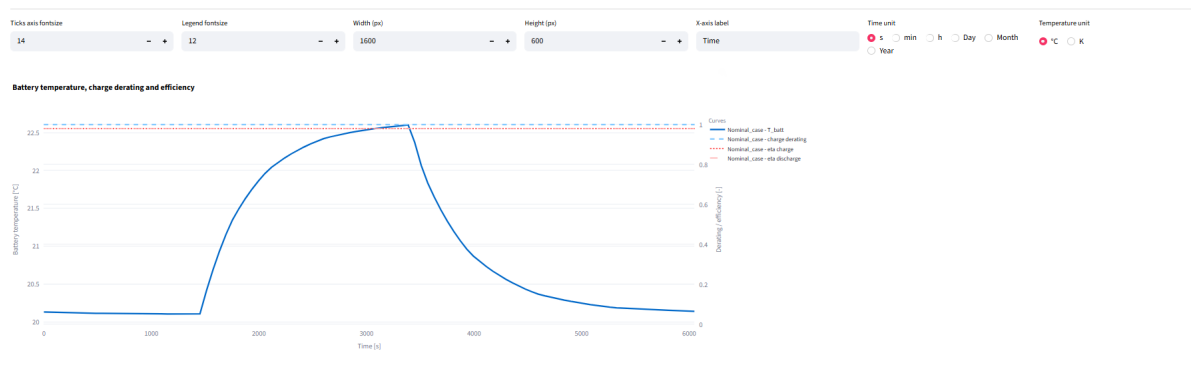


Figure 8.7: Illustrative battery temperature, charge-derating and efficiency histories.

The figures in this subsection demonstrate the form of the available outputs; they do not yet support conclusions about the final nominal, hot or cold cases. Their numerical interpretation requires the validated input parameters and belongs to the results stage of the project.

After the graphs, the battery summary table condenses the complete simulation. It reports final, minimum and maximum SOC; maximum charge and discharge power; curtailed solar energy; unserved-load energy; time spent in CC, CV, solar-limited charge, discharge and voltage-limited discharge; minimum and maximum battery temperature; minimum derating factor; total derated time; and the ranges of charge and discharge efficiency.

A second, optional calculation table retains the values at every output instant. Its principal columns include $Q_{S,elec}$, Q_{load} , direct solar supply, surplus, deficit, charge and discharge powers, terminal and internal voltages, charge and discharge currents, SOC, ϕ , stored energy, efficiencies, curtailed solar power, unserved load, internal and conversion losses, battery temperature, derating factor and operating phase. Together, the graphs and tables connect the selected thermal case with the complete electrical and battery post-processing sequence.

9. Multi-Case Results and Discussion

9.1. Comparison Basis and Verification

9.2. Transient Thermal Comparison

9.2.1. Temperature and Load Evolution

9.2.2. Temperature Changes and Interface Differences

9.2.3. Heat Flows and Integrated Energy

9.3. Heater Influence in the Cold Case

9.4. Electrical and Battery Comparison

9.4.1. Solar Generation and Load Balance

9.4.2. Battery State and Operating Limits

9.5. Main Findings

10. Conclusions and Future Work

10.1. Conclusions

10.2. Future Work

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Application Launch and Interface Overview

A.1. Inspection and Conductor Modules

The model-inspection and conductor-analysis tabs follow the general interface organisation described in Chapter 6. The following views show each area separately so that its main selectors, controls and output regions can be identified clearly.

>> Deploy ⓘ

Flow Explorer: Multilevel Comparison ⓘ

Cases

Choose options

Select at least one case.

Root node

ADCS_UNIT

Total descendants of ADCS_UNIT: 0

Source(s)

ADCS_UNIT

Destination(s)

Space SAT_AFM STRUCTURE BODY BODY_PANEL

Some selected destination children were ignored because their parent group is already selected: STRUCTURE, BODY, BODY_PANEL

Cases

No case selected

Tolerance

1.00

Q Unit

W

mW

Unit

K

°C

The TMD file already contains the temperature unit.

Graph type

nx_global

Time

0.0

Generate graph

Clear graph

Search by name, case, element, notes, type...

Newest first

Saved plots (6)

nx_diagram - Nominal_case - BODY_PANEL_v3

Cases: case SAT_AFM_TRANSISTORIO

Elements: BODY_PANEL

Q: derived from nx_diagram - Nominal_case - BODY_PANEL

nx_diagram - Nominal_case - BODY_PANEL_v2

Figure A.1: Flow Explorer interface for hierarchical heat-flow inspection and multi-case representation.

» Deploy ⓘ

Flow Explorer: Multilevel Comparison **Case info** Group attributes GL conductors GR conductors Nodes Group study TRANSIENT Energy balance

Case info

Cases

Choose options ▼

Select at least one case.

☒ Show Boundary / Inactive node tables

☒ Show node tables with Qi / QR

Select at least one case.

Select a level to view the internal and external coupling summary

Level: 0 ▼ Conductor summary time [s]: 0.0 ▼

Select a level and a group to view their attributes

Level: 0 ▼ Search by name:

Group at selected level: Space ▼

Select at least one case.

Figure A.2: Case-information interface showing the datasets, groups and couplings available in the loaded simulation.

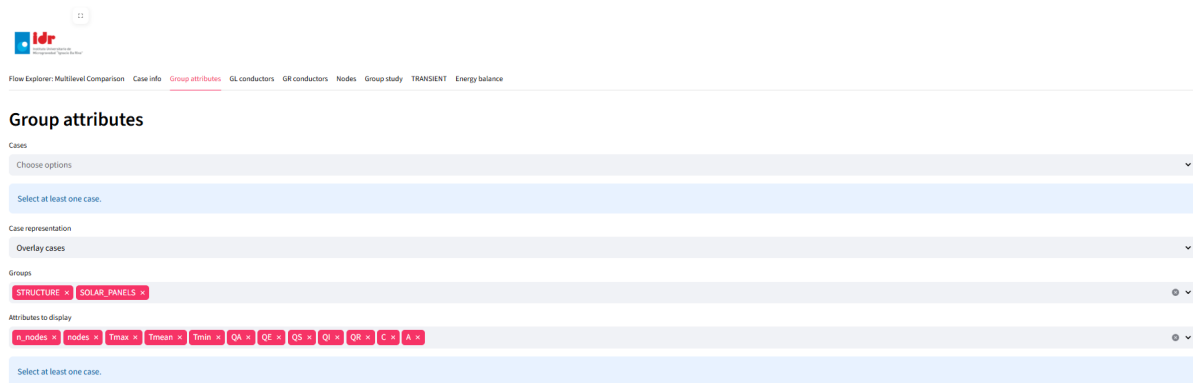


Figure A.3: Group-attribute interface for inspecting the selected model elements at a specified output instant.

» Deploy ⓘ

Flow Explorer: Multilevel Comparison Case info Group attributes GL conductors GR conductors Nodes Group study TRANSIENT Energy balance

GL conductors

Cases

Choose options ▼

Select at least one case.

GL report grouping level: 0 ▼ GL report time [s]: 0.0 ▼ High GL threshold [W/K]: 100 Low GL threshold [W/K]: 0.01

☒ Show detailed GL issue tables

GL table between groups

GL root group: ADCS_UNIT ▼

Total descendants of ADCS_UNIT : 0

Source: ADCS_UNIT ▼ Destination: Space ▼ Time [s]: 0.0 ▼

Saved tables (0)

No saved tables for this section yet.

Generate GL table

Figure A.4: Linear-conductor interface with case, group and output-time controls.

»
Deploy

Flow Explorer: Multilevel Comparison Case info Group attributes GL conductors **GR conductors** Nodes Group study TRANSIENT Energy balance

GR conductors

Cases

Choose options

Select at least one case.

GR report time [s] High GR threshold [m2] Low GR threshold [m2]

0.0 100 0.0001

☒ Show detailed GR issue tables

GR table between groups

GR root group

ADCS_UNIT

Total descendants of ADCS_UNIT: 0

Source Destination Time [s]

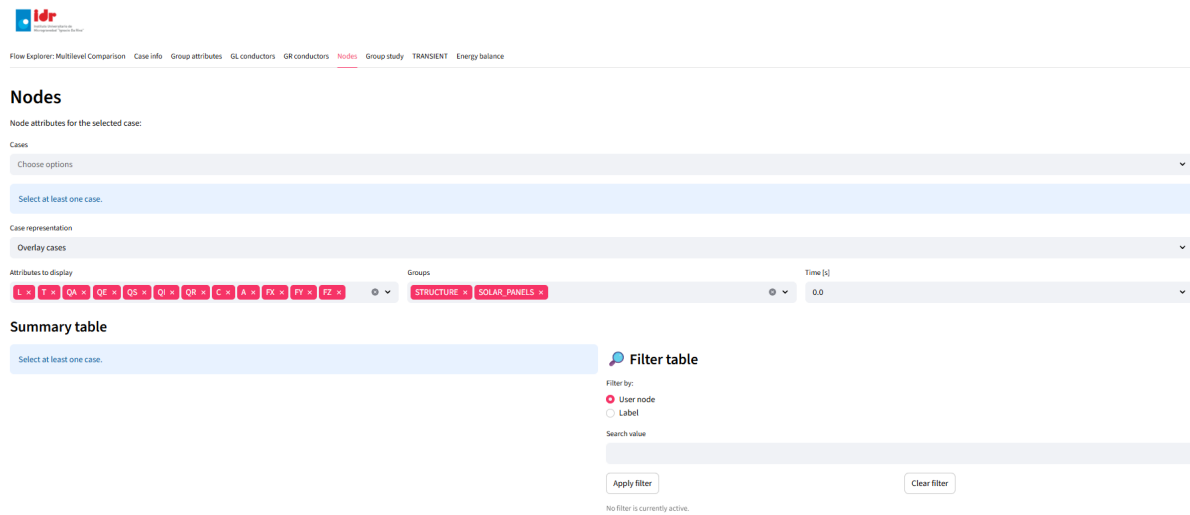
ADCS_UNIT Space 0.0

Saved tables (0)

No saved scenes for this section yet.

Generate GR table

Figure A.5: Radiative-conductor interface with case, group and output-time controls.



Flow Explorer: Multilevel Comparison Case info Group attributes GL conductors GR conductors **Nodes** Group study TRANSIENT Energy balance

Nodes

Node attributes for the selected case:

Cases

Choose options

Select at least one case.

Case representation

Overlay cases

Attributes to display

L T QA QE QS QI QR C A FX FY FZ Groups STRUCTURE SOLAR_PANELS Time [s] 0.0

Summary table

Select at least one case.

Filter table

Filter by:

☒ User node

☐ Label

Search value

Apply filter Clear filter

No filter is currently active.

Figure A.6: Nodal-information interface for the selected hierarchy elements.

Flow Explorer: Multilevel Comparison Case info Group attributes GL conductors GR conductors Nodes **Group study** TRANSIENT Energy balance

Group study

Cases Choose options Group Space Q unit ☒ W ☐ mW

Case representation Overlay cases

Saved tables (0)

No saved scenes for this section yet.

Group conductor analysis

Conductor analysis time [s] 0.0

Figure A.7: Group-study interface for the inspection of selected components and their thermal connections.